A Solver-Independent, Fixed-Topology Linear-Work Endpoint Certificate for Sparse Reciprocal Transportation

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Abstract Large sparse allocation solvers return floating-point candidates whose marginals and optimality are only approximate. This paper develops a solver-independent endpoint certificate for sparse reciprocal transportation, $\min_{z>0,Bz=\beta} \sum_e (c_e z_e + \mu_e/z_e)$ with $\mu_e > 0$, directly on the original sparse arrays. From any finite candidate potential, a reciprocal KKT allocation and canonical spanning-tree correction define an implicit exact-real allocation. A signed, outward-rounded replay either rejects or proves strict positivity, exact feasibility for the represented marginals, and objective gap at most ε. Soundness is independent of the candidate generator under an explicit binary64 and replay-ownership contract. For a fixed candidate, the bound converges to the exact recovery gap as interval widths vanish. With fixed-topology schedules, verification uses $O(m + n) = O(m)$ work and memory; each m-input grouped plan stores exactly $m - g$ additions for g nonempty groups. CertQuota-V makes replay the sole authority over untrusted generators. Evaluation covers certificate sharpness, adversarial degree skew, verifier-only and complete-pipeline scaling, matched-accuracy independently checked conic bounds, dynamic tree policies, hard starts, and budget decisions on historical-only MovieLens workloads. The result is a reusable, structure-specific verified numerical primitive: it avoids an edgewise conic lift, does not trust solver status, and binds the feasibility and gap claims to the same recovered allocation.

Keywords certifying algorithms · interval arithmetic · convex network flow · sparse transportation · primal recovery · reproducible numerical software

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1 Introduction

Sparse convex allocation appears in transportation, resource assignment, and sampling. At scale, a solver ordinarily returns a floating-point point, a residual, and a status flag. None alone establishes that the represented allocation is strictly inside its domain, satisfies its marginals exactly, and lies within a prescribed objective gap. General verified-optimization machinery can address broader models, but a structure-specific endpoint check can be substantially smaller and can expose exactly which original data and floating-point operations carry authority.

We develop such a check for the sparse reciprocal transportation problem

$$P^\star = \inf_{z>0,\ Bz=\beta} P(z), \qquad P(z) = \sum_{e\in E}\left(c_e z_e + \frac{\mu_e}{z_e}\right), \quad \mu_e > 0, \tag{1}$$

where B is an oriented incidence matrix of a connected bipartite graph with $m = |E|$ edges and n vertices. Throughout, c is finite, every $\mu_e > 0$, and $\mathbf{1}^\top\beta = 0$. The global candidate solvers are designed for instances having bipartite-sign marginals and a strictly positive feasible point. The endpoint soundness theorem does not assume an external feasibility oracle: on acceptance, the recovered $Z > 0$ itself discharges nonemptiness for the represented instance.

The setting is deliberately narrower than general convex flow. Newton and interior methods for separable network objectives are classical [37, 38]; specialized solvers exploit bipartite normal equations [11, 58]; and modern theory handles broader flow objectives in almost-linear time [13]. Certifying algorithms and rigorous optimization postprocessing are established paradigms as well [28, 31, 45]. We ask a different endpoint question: can an arbitrary finite candidate potential be checked on the original sparse edge arrays, with no appeal to the generator's status or stopping rule, while preserving support and using only linear work?

The difficulty is not writing a real-arithmetic dual bound. The endpoint must identify one positive exactly feasible primal object, prove the gap for that same object, preserve a prescribed sparse support, and carry every inequality through a finite replay without trusting a conic reduction or solver status. Generic conic verification can represent the problem with one cone block per edge. Earlier network-flow work already recovers primal feasibility from an arbitrary Lagrangian dual vector [41]; candidate independence alone is therefore not our distinction. The contribution here is a smaller original-array, finite-arithmetic contract for this reciprocal structure. Its value is judged as a reusable verification primitive and software interface, not as a new transport objective.

Two identities make the answer possible. First, a dual potential determines an edgewise reciprocal KKT allocation. Second, the residual of that allocation can be routed uniquely over a fixed spanning tree. The resulting correction restores the marginals without introducing an off-support edge, and its exact primal–dual gap is a sum of nonnegative edge terms. Retaining the correction's sign inside an outward-rounded replay turns those identities into a sharp endpoint predicate: one accepted object simultaneously carries positivity, exact-real feasibility, and ε-optimality. Candidate generation and authorization are therefore separate concerns.

One motivating application is variance-aware multi-model federated learning (MMFL). Each client makes one categorical decision among its eligible models, while both client activation budgets and model participation targets are fixed in expectation. Conditional Horvitz–Thompson variance then produces the reciprocal term and the two families of quotas produce the two marginals. This differs from formulations with client capacities and only one global participation total [61]. The application motivates the objective and supplies a conditional-MSE endpoint; it is not needed by the certificate theorem.

The paper answers three questions: what exact-real object an endpoint replay authorizes, how a binary64 implementation encloses that object in linear work, and whether separating a fast generator from strict authority remains useful at sparse-graph scale. Our contributions are:

1. We define a canonical implicit allocation from any finite dual candidate and a support-preserving spanning-tree correction. We prove that signed endpoint acceptance implies strict positivity, exact-real marginal feasibility, and objective gap at most ε, independently of how the candidate was generated.
2. We give an executable original-array replay whose outward intervals bind the domain, tree correction, positivity, and primal–dual identity to the same recovered object; no edgewise conic lift or generator status carries authority.
3. We prove fixed-candidate asymptotic completeness of the signed gap test and bound the irreducible conservatism of its sign-oblivious predecessor. The distinction isolates interval-rounding overhead from loss caused by discarding correction signs.
4. We give a fixed-DAG binary64 realization with explicit rounding and ownership assumptions. On a fixed topology it uses $O(m + n) = O(m)$ work and memory. Each m-input active-frontier grouped reduction stores exactly $m - g$ addition nodes for g nonempty groups, even under arbitrarily skewed degrees. A self-contained content-bound bundle lets a fresh process reconstruct all replay inputs and issue a new authoritative decision. A separate integer/rational checker evaluates the original primal and dual objectives without importing the production verifier.
5. We implement the generate–certify–repair separation in CertQuota-V and evaluate soundness gates, sharpness, adversarial storage, verifier-only and complete-pipeline scaling, dynamic warm starts, named strict pipelines, independently replayed conic candidates, and conditional-MSE sampling on synthetic and real sparse supports. A separately preregistered amendment quantifies tree-choice sensitivity on identical candidates and exercises prospective MovieLens inputs that never inspect post-cutoff ratings. An exploratory revision adds matched-accuracy, independently witnessed conic comparisons and an explicit rational-allocation budget decision interface.

6. As one motivating reduction, we specialize conditional Horvitz–Thompson variance to independent categorical MMFL decisions and obtain the reciprocal objective under two-sided *expected* quotas. This application is not required by the endpoint theorem.

Claim boundary. The new object is the endpoint contract, not the reciprocal objective, Newton method, Laplacian Hessian, tree routing primitive, or general idea of posterior certification. Runtime comparisons apply only to the named implementations, representation, hardware, and warm-start protocol in section 7; they do not rank all transportation or verified-optimization solvers. Candidate-generator independence is a soundness statement about accepted outputs, not a speed claim or an adversarial process-isolation guarantee: a generator may supply arbitrary values and status but may not mutate verifier inputs, code, private capabilities, or floating-point controls during replay. The MMFL studies measure analytic conditional scheduler MSE, not trained-model accuracy, fairness, or time to target.

2 MMFL sampling under two-sided expected quotas

Let $\mathcal{C}$ be clients, $\mathcal{S}$ models, and $E \subseteq \mathcal{C} \times \mathcal{S}$ the sparse eligibility graph. In one round, client i makes one categorical decision: it trains model s with probability $p_{is} > 0$ or is idle. The expected quotas are

$$\sum_{s:(i,s)\in E} p_{is} = r_i, \quad 0 < r_i \le 1, \qquad \sum_{i:(i,s)\in E} p_{is} = d_s > 0, \qquad \sum_i r_i = \sum_s d_s. \tag{2}$$

We assume the sparse support admits a strictly positive allocation $p^\circ > 0$ satisfying these equalities. Thus the row constraint enforces at most one job per client, while d_s is a model participation target only in expectation. Requiring exact integer column counts in every round needs dependent rounding and a different covariance analysis [21].

For a finite represented instance, (r, d) in equation (2) follows the reduced-root convention of section 3: every client quota and every non-root model quota is retained as its exact represented value, while the distinguished model quota is defined by exact balance. An independently supplied binary64 value for that root quota is diagnostic only. Thus every "exact feasibility" statement below refers to this represented quota vector, not to two potentially inconsistent floating-point totals.

Let $u_{is} \in \mathbb{R}^k$ be the prospective local update and $\alpha_{is} \ge 0$ its aggregation weight. To state the oracle objective without anticipating unobserved randomness, define $\mathcal{A}_t := \sigma(\mathcal{F}_t, \{u_{is}, \alpha_{is} : (i,s) \in E\})$. The probabilities p_{is} are $\mathcal{A}_t$-measurable. Conditional on $\mathcal{A}_t$, client decisions are independent across clients and, for every $(i,s) \in E$, $\Pr(Z_{is} = 1 \mid \mathcal{A}_t) = p_{is}$; client i is idle with probability $1 - r_i$. With Z_{is} the selection indicator, define

$$U_s = \sum_{i:(i,s)\in E} \alpha_{is} u_{is}, \qquad \widehat{U}_s = \sum_{i:(i,s)\in E} \frac{Z_{is}\alpha_{is}}{p_{is}} u_{is}. \tag{3}$$

Theorem 2.1 (Conditional Horvitz–Thompson variance identity) *Under the conditional inclusion probabilities and cross-client independence specified above, $\mathbb{E}[\widehat{U}_s \mid \mathcal{A}_t] = U_s$ and*

$$\mathbb{E}\left[\left\|\widehat{U}_s - U_s\right\|_2^2 \mid \mathcal{A}_t\right] = \sum_{i:(i,s)\in E} \alpha_{is}^2 \left\|u_{is}\right\|_2^2 \left(\frac{1}{p_{is}} - 1\right). \tag{4}$$

For model weights $\omega_s > 0$, edge costs $\bar{c}_{is} \geq 0$, and $\gamma \geq 0$, the weighted sum of equation (4) plus $\gamma\mathbb{E}[\sum_{(i,s)\in E} \bar{c}_{is} Z_{is} \mid \mathcal{A}_t]$ differs by a constant from

$$\min_{p>0} \sum_{(i,s)\in E} \left(\frac{\mu_{is}}{p_{is}} + c_{is} p_{is}\right) \quad \textit{s.t. equation (2)}, \tag{5}$$

where $\mu_{is} = \omega_s \alpha_{is}^2 \left\|u_{is}\right\|_2^2$ and $c_{is} = \gamma\bar{c}_{is}$.

The proof is in appendix A. The exact identification with equation (1) requires $\alpha_{is} \left\|u_{is}\right\|_2 > 0$ on every declared edge. We therefore define the studied instance only after choosing a support on which $\mu_{is} > 0$. A zero-weight eligibility relation lies outside that instance class: removing it is an application-level change of support that can alter the feasible set, and replacing it by a positive floor defines a surrogate objective. Neither operation is an exact numerical preprocessing rule for equation (1).

Exact update norms are also not normally known before scheduling. We therefore distinguish a nondeployable current-norm oracle from stale-norm and coordinate proxies; a proxy allocation is optimal only for the surrogate weights supplied to it. This statistical limitation is separate from the numerical validity of an accepted allocation. A second distinction concerns representation: the endpoint certificate below binds an implicit exact-real allocation, whereas a conditional-MSE value computed from an emitted binary64 materialization belongs to that finite vector. The latter must be used consistently as both the categorical inclusion law and the Horvitz–Thompson denominator; certificate acceptance alone does not verify a sampler implementation.

3 Dual candidate and exact-real tree recovery

Orient every edge from the client side to the model side, so $Bz = \beta = (r, -d)$. For the theorem, let v_0 be a distinguished root. The mathematical input is the $n - 1$ vector of independent binary64 quotas $\bar{\beta}_{V\setminus\{v_0\}}$. It retains those components as exact dyadics and defines the root in exact real arithmetic by

$$\beta_v := \bar{\beta}_v \ (v \neq v_0), \qquad \beta_{v_0} := -\sum_{v\neq v_0} \bar{\beta}_v, \qquad \varDelta_{\mathrm{root}} := \beta_{v_0} - \bar{\beta}_{v_0}. \tag{6}$$

The recommended public constructor accepts only those $n-1$ independent values and derives the root. For backward compatibility, the full-vector constructor fixes $v_0 = n-1$, rejects nonfinite entries and a declared imbalance larger than $10^{-9}\max\{1, \|\bar{\beta}\|_1\}$, then stores the supplied root, an `fsum`-based binary64 representative of the implied root, and a rounded adjustment diagnostic $\widehat{\Delta}_{\mathrm{root}}$. That compatibility value is not an independent constraint. The exact-real Δ_{root} in equation (6) is a mathematical definition reconstructed from the independent entries during strict replay; it is not the stored diagnostic. The constructor does not decide general sparse transportation feasibility. Experimental generators numerically seed quotas from a positive binary64 planted flow, but its rounded marginals are not asserted to equal Bx in exact-real arithmetic after the represented-root convention is imposed. For every reported accepted instance, endpoint recovery itself supplies a strictly positive exactly feasible $Z(P, y, T)$. Other violations may lead only to input rejection, certificate rejection, or explicit solver failure, never to a strict accepted allocation. All strict claims refer to β in equation (6), not to the independently rounded vector $\bar{\beta}$. Operationally, the strict replay need not materialize the generally non-binary64 root quota: it computes every non-root residual interval and defines the root residual interval as the directed negative sum of those intervals. This is algebraically equivalent to the exact-real root convention.

For any stored candidate potential y, interpreted as an exact dyadic vector, set

$$q(y) = c - B^\top y, \qquad x_e(y) = \sqrt{\mu_e / q_e(y)}, \qquad g(y) = Bx(y) - \beta. \tag{7}$$

When $q > 0$, a dual lower bound is

$$D(y) = \beta^\top y + 2\sum_e \sqrt{\mu_e q_e(y)}. \tag{8}$$

Fix a spanning tree T. Because g is exactly balanced under equation (6), there is a unique tree-supported flow $h_T(y)$ satisfying $Bh_T(y) = -g(y)$. Computing a balanced flow on a tree is classical [1]; our use combines that routing primitive with the reciprocal primal–dual identity and an executable outward-rounded certificate. Define the canonical recovered object

$$Z(P, y, T) := x(y) + h_T(y). \tag{9}$$

This is an *implicit exact-real allocation*. It is represented by the problem, candidate, tree, and certificate. A downstream binary64 array $\widehat{z}$ materializes the same formula numerically, but it is not asserted to satisfy $B\widehat{z} = \beta$ bit for bit; its measured marginal residual is always reported.

4 Candidate-generator-independent endpoint verification

The authoritative verifier uses outward rounding to prove: (i) $q_e > 0$; (ii) lower bounds $\ell_e \le x_e(y)$; (iii) signed enclosures $H_e = [\underline{h}_e, \overline{h}_e] \ni h_{T,e}(y)$; (iv) lower bounds

$$\underline{z}_e := \mathrm{down}(\ell_e + \underline{h}_e) > 0 \quad (e \in T)$$

on the recovered allocation; and (v), with $a_e := \max\{|\underline{h}_e|, |\overline{h}_e|\}$,

$$\Gamma_U := \mathrm{up} \sum_{e \in T} \frac{\mu_e a_e^2}{\ell_e^2 \underline{z}_e} \leq \varepsilon. \tag{10}$$

Non-tree correction coordinates are exactly zero. The operation up denotes an outward upper evaluation of the complete reduction DAG, not merely a final rounded sum.

EndpointVerify(P, y, T, ε). The authoritative routine is the following fixed decision procedure.

1. Validate dimensions, finite inputs, $\mu > 0$, the root convention, and that T spans the declared graph; otherwise do not authorize a return. The low-level API may raise on malformed input.
2. Evaluate outward intervals for every slack q_e; reject unless each lower endpoint is positive. Evaluate intervals for x and for the grouped residual $g = Bx - \beta$ using the precompiled reduction DAGs.
3. Route the residual interval over T to enclose the unique tree flow f_T satisfying $Bf_T = g$, retain the signed enclosure $H_e = -F_e$ for $h_T = -f_T$, and reject unless every outward lower bound $\underline{z}_e = \mathrm{down}(\ell_e + \underline{h}_e)$ is positive.
4. Evaluate the complete outward gap bound equation (10). Return accept exactly when $\Gamma_U \leq \varepsilon$; otherwise return reject. The logical acceptance trace contains $(P, y, T, \varepsilon, \Gamma_U, \{\ell_e, \underline{h}_e, \overline{h}_e, \underline{z}_e\})$. The software returns an immutable replay key and compact summary rather than retaining both full edgewise interval arrays.

Every exceptional arithmetic event is a rejection. A binary64 materialization may accompany an accepted tuple, but it is not the object authorized by this procedure. SolveVerifyRepair catches low-level exceptions and records an explicit failed attempt with no allocation; the compact result does not retain a primitive-level rejection reason code. Appendix C.1 gives the primitive-level replay.

Theorem 4.1 (Solver-independent endpoint soundness) *Let P be a validated represented instance of equation (1): its declared graph is connected, its arrays are finite with $\mu_e > 0$, and its marginals use the reduced-root convention equation (6). Let T be a validated spanning tree of that graph, let $0 < \varepsilon < +\infty$, and let y be any finite candidate array. Under the execution contract in theorem C.1, if the authoritative verifier accepts (P, y, T, ε), then $Z(P, y, T)$ is strictly positive, $BZ(P, y, T) = \beta$ in exact-real arithmetic, and*

$$P(Z(P, y, T)) - P^\star \leq \Gamma_U \leq \varepsilon. \tag{11}$$

The conclusion is independent of how accurately, deterministically, or successfully y was generated, subject to the replay ownership and arithmetic contract.

Proof The executable-to-real bridge is lemma C.1: on every accepting replay, each interval used below contains its exact-real DAG value. The domain bounds therefore make $x(y)$ positive. On a tree edge, $x_e + h_e \geq \ell_e + \underline{h}_e \geq \underline{z}_e > 0$; non-tree entries remain $x_e > 0$. By construction, $B(x + h_T) = \beta + g - g = \beta$. Thus acceptance itself supplies a strictly positive feasible point. In particular, for every left vertex i,

$$\beta_i = \sum_{e \ni i} Z_e > 0.$$

Consequently every $\widetilde{z} \geq 0$ with $B\widetilde{z} = \beta$ satisfies $0 \leq \widetilde{z}_{is} \leq \beta_i$; the nonnegative feasible closure is therefore nonempty and compact. Extending P by $+\infty$ whenever an edge coordinate is zero, the reciprocal barrier gives attainment in the positive interior, and strict convexity gives uniqueness. For every positive feasible u, coordinatewise minimization of the Lagrangian gives $P(u) \geq D(y)$, so in particular $P^\star \geq D(y)$. For $z = x + h_T$, feasibility and $q_e = \mu_e / x_e^2$ give

$$P(z) - D(y) = \sum_e \left(q_e z_e + \frac{\mu_e}{z_e} - 2\sqrt{\mu_e q_e} \right) \tag{12}$$

$$= \sum_e \frac{\mu_e h_e^2}{x_e^2 (x_e + h_e)}. \tag{13}$$

Weak duality and the certified inequalities bound each nonzero term by the corresponding summand of equation (10): its numerator contains h_e^2, while $x_e \geq \ell_e$ and $x_e + h_e \geq \underline{z}_e$. The outward reduction contains their sum. No step uses how y was obtained.

Proposition 4.1 (Fixed-DAG interval continuity and fixed-candidate completeness) *Fix (P, y, T) with $q(y) > 0$ and $Z(P, y, T) > 0$. Consider interval evaluations of the same finite expression DAG, indexed by k. For every exact-real node value v, require its enclosure $I_{k,v}$ to satisfy $v \in I_{k,v}$ and $\operatorname{diam}(I_{k,v}) \to 0$. For all sufficiently large k, the domain and positivity denominators are positive; let $\mathit{\Gamma}_{U,k}$ denote the resulting upper gap endpoint. Then $\mathit{\Gamma}_{U,k} \geq G(y, T)$ for every such k and $\mathit{\Gamma}_{U,k} \to G(y, T)$, where*

$$G(y, T) := P(Z(P, y, T)) - D(y) = \sum_{e \in T} \frac{\mu_e h_e^2}{x_e^2 (x_e + h_e)}. \tag{14}$$

Consequently, if $G(y, T) < \varepsilon$, every sufficiently narrow member of this interval-evaluation family accepts. A fixed binary64 replay remains deliberately fail-closed and may reject such a candidate. No nesting or monotonic decrease of the interval endpoints is assumed.

The sign retention is essential for this conclusion. The legacy sign-oblivious predicate first requires $x_e - |h_e| > 0$ on every tree edge. If $\rho := \max_{e \in T} |h_e| / x_e \geq 1$, that positivity gate rejects (equivalently, set $\mathit{\Gamma}_{\mathrm{abs}} = +\infty$). Only when $\rho < 1$ does its limiting bound equal

$$\mathit{\Gamma}_{\mathrm{abs}}(y, T) := \sum_{e \in T} \frac{\mu_e h_e^2}{x_e^2 (x_e - |h_e|)}. \tag{15}$$

and then

$$G(y,T) \le \mathit{\Gamma}_{\mathrm{abs}}(y,T) \le \frac{1+\rho}{1-\rho}\, G(y,T). \tag{16}$$

Thus the old bound is asymptotically sharp as $\rho \to 0$, but it can remain strictly conservative at any fixed nonzero correction. Proofs of both claims are in appendix C.

Corollary 4.1 (Generate–certify–repair) *Consider any finite or adaptive sequence of candidate generators, optionally ending with any fallback generator whose candidate is replayed by* EndpointVerify*. A wrapper that returns only the first endpoint-accepted candidate returns an ε-optimal implicit allocation. Generator failure and verifier false rejection affect completeness and runtime, not soundness.*

4.1 Linear endpoint replay

For a fixed graph and tree, we precompile stable client and model grouped-sum schedules, balanced pairwise scalar-reduction DAGs, and each tree level's active parents and child groups. Numeric interval endpoints are recomputed for every candidate and never reused across iterates. A verification then uses $O(m)$ slack and square-root interval operations, $O(m+n)$ grouped accumulation, $O(n)$ tree routing, and $O(n)$ positivity, gap, and scalar-reduction work. For any one grouped reduction plan with m inputs partitioned into nonempty groups of degrees $d_1, \ldots, d_g > 0$, the active-frontier compiler creates exactly $\sum_j (d_j - 1) = m - g$ addition nodes: a completed singleton leaves the frontier rather than being copied through the remaining levels. The count is per grouped plan, not the total over all incidence, tree-level, and scalar plans. Applied plan by plan, it rules out a hidden $m \log d_{\max}$ storage term on skewed topologies.

Theorem 4.2 (Linear endpoint replay) *On a fixed connected topology with precompiled schedules, endpoint verification uses $O(m+n) = O(m)$ arithmetic work and $O(m+n)$ memory under the fixed-width binary64 unit-cost arithmetic/word model. Schedule construction, candidate generation, and optional arbitrary-precision replay are outside this bound.*

For context, appendix D also records a restricted fresh-array probe observation: an accepting edgewise dual-slack verifier must inspect every slack-sensitive cost word. This is not a lower bound for arbitrary certificate systems or candidate generation.

5 CertQuota-V: fast candidates, strict authority

SolveVerifyRepair(P, T, ε). Given an ordered, finite list of untrusted candidate generators $(\mathcal{G}_1, \ldots, \mathcal{G}_K)$, the wrapper executes:

1. for $k = 1, \ldots, K$, request a candidate y_k and retain the generator status and diagnostics;
2. if generation fails or y_k is nonfinite, retain an explicit failed attempt and continue;
3. call $\textsc{EndpointVerify}(P, y_k, T, \varepsilon)$ and return only its first accepted implicit allocation and certificate;
4. if all attempts reject or fail, return an explicit failure record with the complete attempt log and no allocation.

Thus the wrapper is complete only to the extent that one listed generator enters the accepted set, while its returned-object soundness is independent of every generator.

CertQuota-V separates generation from authority. Its frozen implementation uses the following cascade:

1. run damped Newton with a matrix-free Jacobi preconditioner to target $\varepsilon/16$, then perform an independent endpoint replay;
2. after rejection, run a tighter PyAMG candidate [5] to target $\varepsilon/256$ and replay the verifier;
3. when the reduced dimension is at most 5,000, try a sparse-direct repair;
4. try reciprocal alternate scaling; and
5. invoke CertQuota-S, the interval-screened Newton candidate generator, as the final repair/audit mode; its return still requires a fresh terminal endpoint replay.

Every branch is an untrusted proposal. Sparse-direct, alternate-scaling, and fallback status codes cannot bypass theorem 4.1. An invalid or rejected candidate produces another attempt or an explicit failure record; it never produces a fabricated allocation.

The common fast case has one candidate solve, one recovery-only interval state, one endpoint certificate, and one binary64 materialization. The returned strict object remains $Z(P, y, T)$ in equation (9); materialization time and $\|B\widehat{z} - \beta\|_1$ are separate fields. The authority contract is candidate-generator-independent: IEEE-V evaluates the endpoint DAG with directed binary64 rounding, while Arb-V evaluates the same mathematical test using 70-decimal-digit Arb ball arithmetic [33]. Accordingly, experiments report strict-certificate coverage for the implicit Z separately from conditional MSE and marginal residuals evaluated at $\widehat{z}$; the former does not turn the latter into an interval-certified or exactly quota-feasible probability vector.

5.1 Candidate-path diagnostics are auxiliary

The endpoint theorem says nothing about intermediate iterates or global convergence of a candidate generator. CertQuota-S uses the Laplacian Hessian $H = BWB^\top$ and conservative tree-energy bounds to screen a local Newton basin and inexact-direction residuals. The exact-real recurrences, constants, and proofs are relegated to appendix E, together with the precise gap between the mathematical transition and its rounded implementation.

These screens are useful for choosing a fallback, but they are intentionally not part of the endpoint contract. A stored binary64 update may differ from the exact-real transition used by the local analysis; neither a basin screen, a forcing screen, nor a solver status authorizes the result. Every candidate, including a CertQuota-S candidate, must pass a fresh terminal replay under theorem 4.1. Keeping this material in the appendix separates the paper's primary contribution—a finite endpoint decision—from an optional candidate-generation path.

6 Finite-precision implementation

Arithmetic contract. Stored binary64 inputs are interpreted as exact dyadic numbers. Directed rounding encloses subtraction, square root, grouped incidence products, tree flows, positivity differences, and every term and reduction in equation (10). The root convention equation (6) is evaluated from the independent entries. The soundness lemma assumes round-to-nearest binary64, gradual underflow, and correctly rounded primitive operations and square root. These semantics follow the declared IEEE 754 binary64 model [27]; they are assumptions of the theorem rather than facts inferred from a solver status. The timed IEEE path fails closed unless structured probes pass for the binary64 layout, `nextafter` boundaries, ties-to-even, subnormal arithmetic, selected four-operation identities, and selected square roots. It rechecks current-thread control sentinels at every public certificate entry, including prepared-state reuse. These probes detect important violations but do not formally establish the assumptions over every input or future execution path. The IEEE path contains no hidden Arb call; Arb is a separately implemented audit backend, not a fully independent external baseline.

Topology versus numeric state. Only immutable topology schedules are cached. We content-validate a common topology owner for dynamic problems with identical endpoints, prewarm the shared problem and tree plans outside method timing, and separately report cold-plan construction. Stepwise code whose cache is keyed by current problem identity is also prewarmed before the rotated method block. Numeric slack, root, residual, tree-flow, and gap intervals are freshly computed for every candidate.

Portable third-party replay. An in-process certificate summary is capability-bound and deliberately loses authority when serialized. The public `certquota-endpoint-replay-v1` bundle instead stores the exact binary64 problem arrays (including the declared compatibility root), candidate, tree, tolerance, backend request, and a canonical SHA-256 payload binding. The installed `certquota-verify` command reconstructs new immutable problem and router objects, checks their fingerprints, and reruns the strict backend. Only the newly issued certificate can authorize acceptance; the stored summary is used solely to detect decision disagreement. Thus a downstream reviewer can carry and replay the witness without trusting a pickle, Python object identity, or the original process.

Independent conic candidate. For small and medium references we introduce epigraph variables $x_e, t_e > 0$ with $x_e t_e \geq 1$, represented by the standard cone

$$\|(2, x_e - t_e)\|_2 \leq x_e + t_e, \qquad Bx = \beta, \qquad \min c^\top x + \mu^\top t. \tag{17}$$

CVXPY [19] constructs equation (17), and Clarabel [23] supplies an equality-dual candidate. Objective and row scaling, including the Lagrangian sign, are undone to recover the canonical y before the same IEEE endpoint verifier is applied. The derivation is given in appendix F.

7 Experimental protocol

Evidence authority. The principal results belong to the clean A42 confirmatory execution. A source manifest is created once after code and protocol freeze and before any formal producer runs; each raw file records its digest. Earlier development releases remain immutable, and no prior raw row, timing, aggregate, bootstrap draw, figure, or table is pooled into A42. Analyses fail closed on source or raw-file hashes, schemas, numeric integrity, planned-cell coverage, and certificate authority. Manuscript numbers and figures are generated from the twelve validated analyses, and a final read-only audit recomputes those analyses, rebuilds every figure, runs the full tests and package checks, and compiles the paper from an isolated source copy. The exact closure and commands are recorded in appendix H.

A43 is a disjoint preregistered reviewer-obstacle amendment, not an extension of the A42 population. Its source manifest binds 139 executable and protocol files before either formal panel runs. It retains all four tree choices, including negative outcomes, and derives the historical MovieLens workload only from rows at or before each cutoff. Its raw records, summaries, portable replay, and deterministic recomputation are bound separately; no A43 row is used in an A42 estimate or confidence interval.

Timing and accuracy rules. Runs use a single-thread process contract on a 12-core AMD Ryzen 9 5900X workstation with 64 GiB RAM, Python 3.10, NumPy 1.26.4, SciPy 1.11.4, PyAMG 5.2.1, CVXPY 1.6.5, and python-flint 0.9.0. Candidate or method order rotates within paired rounds. In method comparisons, topology construction and plan prewarming are excluded; complete-pipeline timing begins immediately before `solve_verified` and ends after binary64 materialization. Cold topology and plan construction are reported separately. Accuracy requires positivity, normalized marginal ℓ_1 residual at most 10^{-9}, relative objective gap at most 10^{-8} when a planted reference is available, and, for every strict method, a terminal endpoint acceptance. Confidence intervals use 10,000 graph-seed or sequence-cluster bootstrap resamples as appropriate. Fitted runtime exponents are descriptive and are not substitutes for theorem 4.2.

Evaluation questions and panels. The frozen panels address five distinct questions:

1. *Does the predicate fail closed, and where is it conservative?* Fifty-four behavior cases cover direct acceptance, repair after rejection, and domain-invalid candidates. A separate high-precision sharpness panel compares the exact recovery gap, the limiting sign-oblivious bound, the signed binary64 bound, and the legacy global-ℓ_1 decision over frozen graph, perturbation, and tolerance grids.
2. *Do the claimed storage and work survive sparse-graph extremes?* An adversarial grouped-reduction panel combines one giant group with singleton groups through ten million inputs and checks the exact $m - g$ node count, reference containment, and a linear byte bound. Verifier-only scaling runs from 30,000 to ten million edges. Complete solve–verify–materialize scaling uses three graph families, three seeds, and five sizes through three million edges, for 45 setup cells and 135 timed measurements after excluded warm-ups.
3. *What does strict authority cost across generators?* The study contains 1,050 paired dynamic rounds at 100,000 edges, 75 paired rounds for each of four named strict pipelines, a 45-cell internal conic reference, and an 18-cell external conic stress replay. A standalone 256-bit Arb implementation replays conic candidate extraction, Kruskal tree repair, and the primal–dual gap without importing the production solver, verifier, interval, or tree modules; designated rejection cases receive an additional 384-bit replay. This is a numerical independence audit, not a generic verified-conic baseline or a timing ranking.
4. *Does the reciprocal objective express a meaningful sampling tradeoff?* A synthetic panel crosses 32 sparse graph cells, five independent graph/update replicates, 20 burn-in fields, 20 evaluation fields, and 11 policies. Two MovieLens-1M panels use the official interaction archive: one keeps its user–movie topology with planted numerical quantities, and one constructs a user–genre eligibility graph from temporal rating summaries and compares current, historical, and equal-weight policies under equal and popularity-aware expected quotas.
5. *How much does tree choice matter, and can the public-data workload be defined prospectively?* A43 replays identical candidates through input-order, maximum-conductance, minimum-conductance, and seeded-random trees over 144 candidate–tolerance cells. A separate MovieLens panel uses cutoff quantiles 0.40, 0.50, and 0.60 and constructs support, μ, quotas, start, and tree without reading later ratings. One accepted endpoint is exported as a canonical replay bundle and checked in a fresh reconstruction.

The sampling endpoint is the analytic conditional MSE in theorem 2.1; the formal artifact records `monte_carlo_rounds`=0. The MovieLens archive checksum and numeric seeds are fixed, no demographic fields are read, and the data are neither extracted nor redistributed. The user–movie panel makes no recommendation-effect claim, while the user–genre study is a descriptive conditional-MSE case study rather than a neural-training or ranking evaluation.

8 Results

We organize the evidence around five questions: whether the endpoint authority fails closed and remains sharp; whether its implementation realizes the proved linear structure; what strict authority costs in complete dynamic pipelines; whether the resulting allocations improve the conditional-MSE objective for materialized probabilities; and how tree choice and prospectively constructed public-data inputs affect acceptance. Unless stated otherwise, timings are single-thread wall times on the registered machine. Scaling exponents and method timings are descriptive implementation measurements, not substitutes for theorem 4.2 or universal solver rankings.

8.1 Fail-closed authority and certificate sharpness

The behavior experiment presents 54 candidates spanning direct acceptance, repairable rejection, and invalid-domain failure. The verifier accepts 18 directly, rejects and repairs 18 to strict endpoints, and returns explicit failure with no allocation for all 18 invalid-domain cases. Hence all 36 returned allocations have valid certificates, while no invalid candidate escapes the fail-closed path (figure 1).

A separate sharpness panel uses 210 fixed candidates across three graph families, three scales, and seven perturbation radii. Exact high-precision replay finds all 210 recovered allocations positive, and the edgewise signed predicate proves positivity in all 210 cases; the global ℓ_1 screen proves only 158. Across relative tolerances $10^{-6}, \ldots, 10^{-14}$, neither predicate has a false accept. Whenever the exact recovery gap is below a tested tolerance, the production signed predicate also accepts, giving 100% recall on the applicable fixed-candidate cells. At tolerance 10^{-10} alone, the edgewise predicate authorizes 49 candidates rejected by the global screen.

The median signed IEEE bound divided by the exact recovery gap is 1.00000072, whereas the median global-screen inflation is `1.89906e+07`. The exact-real structural bound has median inflation 1.00000097, and its nonzero-radius excess has log–log slope 0.9883 as the perturbation shrinks. Signed cancellation does not change the accept/reject count on this grid, but it can reduce the same-enclosure absolute-correction bound to 0.112 of its value. Zero-radius cells expose an unavoidable floating-point enclosure floor and

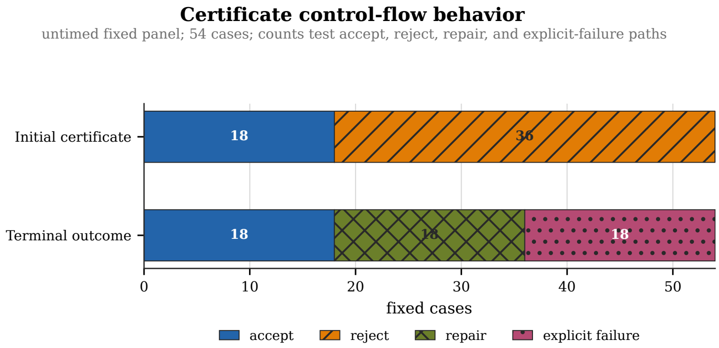

Fig. 1 A42 fail-closed behavior. An initial rejection can lead to a separately verified repair; domain-invalid inputs produce no allocation.

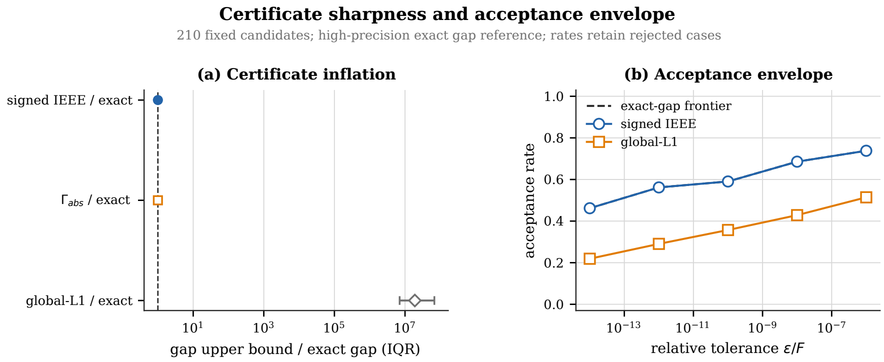

Fig. 2 Certificate sharpness against high-precision exact-recovery quantities. The edgewise signed construction retains local correction information that the global ℓ_1 screen discards.

are reported separately in the artifact. These observations support the fixed-candidate sharpness mechanism in proposition 4.1; they do not establish completeness for arbitrary binary64 candidates.

8.2 Linear data structures, verifier scaling, and complete-pipeline scaling

The exact $m - g$ addition count matters on highly irregular incidence arrays. We therefore stress the grouped-reduction compiler with one giant group and many singleton groups at sizes from 30,000 to ten million. All 60 cells satisfy the node-count, workspace, storage, execution, and enclosure checks. The 30 cells through 300,000 inputs also bitwise-match the independently executed levelwise reference; larger cells use the registered analytic/high-precision enclosure. The plan uses 10 index bytes per input, below its 12-byte registered bound, and the descriptive time slope is 1.01528. In particular, completed singleton groups leave the active frontier instead of being carried through later levels (figure 3).

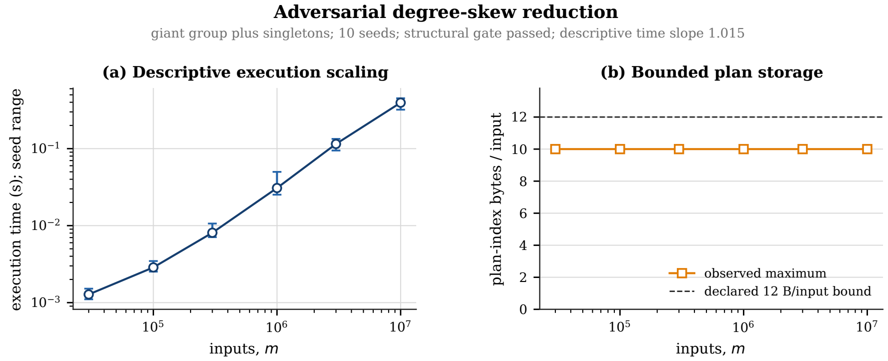

Fig. 3 Adversarial segmented-reduction stress. Exact plan storage stays linear under the one-giant-group-plus-singletons degree pattern.

Table 1 A42 complete solve–verify–materialize scaling. Each size aggregates nine graph-family–seed medians; all 135 timed measurements are endpoint-accepted.

Edges	GM complete-pipeline seconds
30,000	0.06005
100,000	0.15190
300,000	0.92186
1,000,000	3.05713
3,000,000	9.31139

Verifier-only scaling contains 60 graph–seed cells and 180 timed replays. The seed-clustered exponent is 1.02575 (95% interval $[1.01259, 1.03783]$). The interval excludes one, so we describe the measurement as near-linear rather than claiming an exact empirical exponent. The largest instance has 10000000 edges, a geometric-mean replay time of 3.182 seconds, and 51.6079 measured topology-plan bytes per edge. Cold problem- and tree-plan construction is recorded but excluded from this verifier-only timing.

The complete pipeline adds candidate generation, strict-state preparation, endpoint verification, fallback accounting, and materialization. It contains 180 records: 45 setup/warmup cells and 135 timed measurements across three graph families. Every timed measurement returns a strict endpoint. Table 1 shows the graph-family–seed median geometric means. At three million edges the complete time is 9.31139 seconds. Its descriptive slope, 1.13588, is higher than the verifier-only slope; candidate generation consumes a mean fraction 0.79933 of wall time, versus 0.0478034 for verification. This phase profile is consistent with, but does not by itself explain, the slope difference. The pipeline slope was explicitly not a publication gate. Here “complete” means the public `solve_verified` call through returned materialization after cold graph, router, and topology-plan setup; it is not first-use end-to-end latency.

Together, the skew, verifier-only, and complete-pipeline panels distinguish three claims that are often conflated: exact linear plan size, near-linear observed

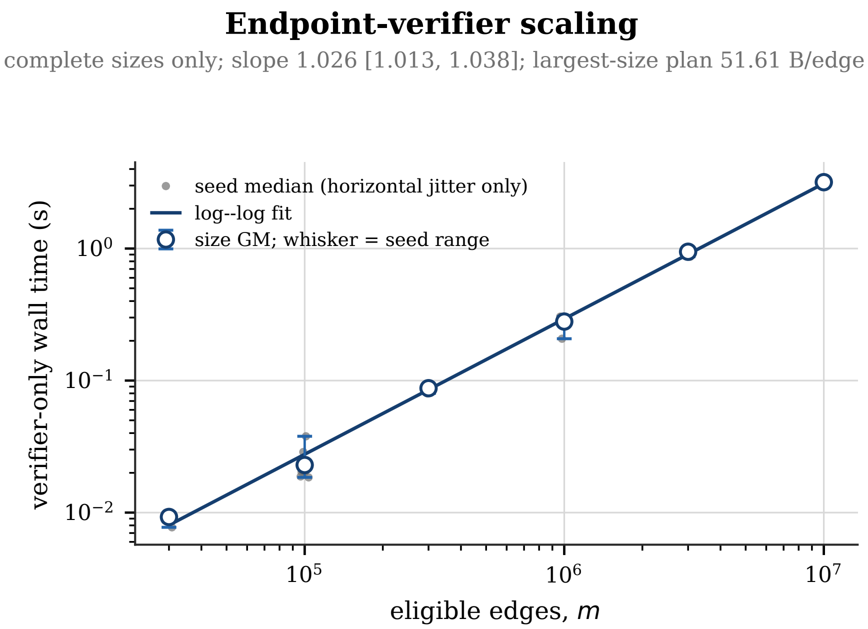

Fig. 4 Verifier-only scaling through ten million edges. Points are graph-seed medians; the fitted interval clusters by seed.

verifier time, and the larger cost of whatever candidate generator is placed in front of the verifier.

8.3 Dynamic and strict-pipeline cost

The dynamic panel evaluates 1,050 paired rounds at 100,000 edges. CertQuota-V returns a strict, matched endpoint on every round with geometric-mean complete time 0.105565 seconds. The interval-screened stepwise strict pipeline takes 0.362664 seconds, so the paired V/stepwise ratio is 0.291082 (95% interval $[0.28449, 0.29793]$). Uncertified damped Newton is faster at 0.0722185 seconds: V/damped is 1.46175 (95% interval $[1.4354, 1.4884]$). The latter returns numerically matched points but supplies no strict endpoint authority, and is therefore an overhead reference rather than a certified competitor.

The warm-start basin screen passes only 720 of 1050 rounds: all 300 smooth 10^{-3} updates and 420 non-jump shock-sequence rounds, but neither the 300 smooth 10^{-2} updates nor the 30 shock jumps. Nevertheless, the final endpoint verifier accepts 1050/1050 CertQuota-V outputs. Thus basin membership is a useful local diagnostic, not the authority for the returned allocation; the solve–verify cascade can recover outside that diagnostic region.

All four named strict pipelines also certify and match each of 75 frozen 100,000-edge rounds. Table 2 reports their descriptive cost. CertQuota-V is the

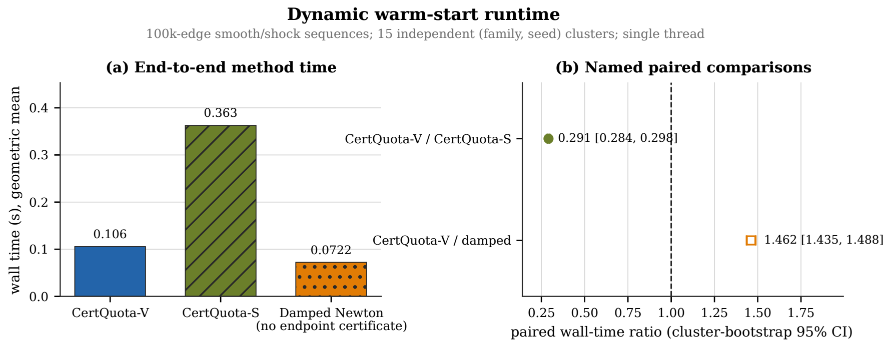

Fig. 5 Complete dynamic time and paired ratios. The dashed line marks equal time; damped Newton is deliberately shown as an uncertified numerical reference.

Table 2 Named strict pipelines at 100,000 edges. Ratios are paired method time divided by CertQuota-V time.

Strict pipeline	GM seconds	Method/V
CertQuota-V	0.110055	1.000
Stepwise interval screen	0.400204	3.636
Alternate-scaling verified	1.13563	10.3187
Arb terminal replay	2.34512	21.309

lowest-time implementation in this panel, but the other three reuse portions of our recovery or certification machinery, so the panel cannot support a general fastest-solver claim. Alternate scaling is especially long-tailed: its median is 0.454156 seconds, while its 90th and 95th percentiles are 139.547 and 141.645 seconds and its maximum is 147.939 seconds.

The conic experiments supply a more independent candidate source. On 45 small/medium internal cells, both CertQuota-V and Clarabel/CVXPY candidates receive endpoint certificates. On the 18-cell external stress grid, CertQuota-V certifies 18/18, whereas Clarabel reports an optimal candidate on every cell but only 14/18 pass endpoint replay. Consequently no strict external timing ratio is reported. A standalone Arb implementation, sharing neither the production interval code nor tree router, reproduces all 18 decisions at 256 bits and the four rejections again at 384 bits. In the rejected cells, its same-dual gap intervals lie wholly above the respective targets: lower endpoints $0.0532577, 0.0499891, 0.0382662, 0.0302070$ versus targets $0.0254701, 0.0262432, 0.0262347, 0.0256089$. The production upper bounds exceed the independent values by at most 9.19×10^{-8} percent. Arb therefore diagnoses insufficient accuracy of the submitted dual witnesses, rather than rounding conservatism in the production certificate. Recovered primals do pass against the planted reference dual, so the diagnosis does not imply that the raw primal coordinates are themselves unusable.

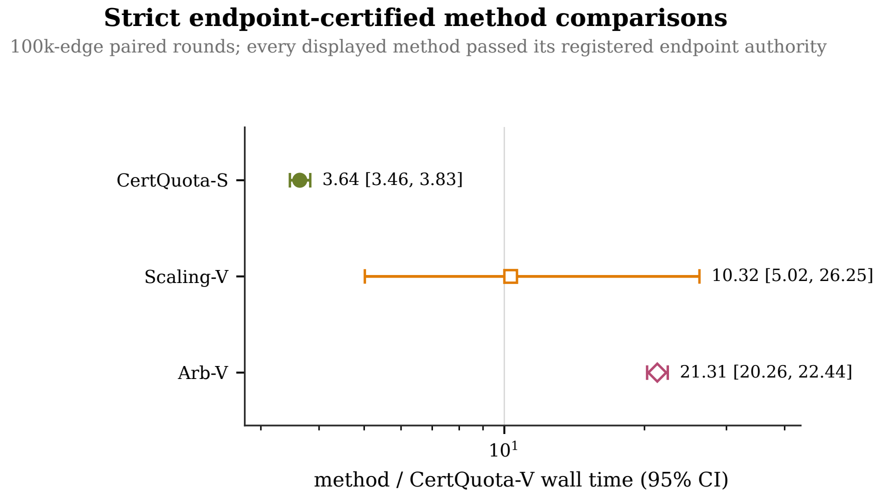

Fig. 6 Descriptive ratios for the four named strict pipelines. Their shared components limit the comparison to these executed implementations.

8.4 Materialized conditional-MSE case studies

The synthetic sampling panel contains 3,200 quota-matched analytic records per policy and 32,000 nonuniform solves. All 32,000 emitted probability vectors have strict endpoint certificates; 20 attempted candidates are rejected and 10 records are repaired by a later cascade stage. The maximum certified objective-gap upper bound is 1.07×10^{-12} and the maximum measured normalized marginal residual after binary64 materialization is 9.24×10^{-13}. The registered row targets obey $\max_i r_i \leq 0.2$; together with this residual bound, every materialized row mass remains strictly below one. The analytic MSE uses that same positive finite vector both as the categorical inclusion law and as the Horvitz–Thompson denominator. The current-norm and age-one policies pass the two registered Holm-adjusted one-sided tests ($p_{\text{adj}} = 2.00 \times 10^{-4}$).

The aggregate effect requires quota stratification (table 3). With quota ratio one, the current oracle improves over quota-matched uniform sampling by only about 0.34%; at ratio ten, the improvement is about 14.5%. An age-one proxy closely tracks the oracle in both regimes, while age 20 loses some benefit. The coordinate proxy is harmful in both regimes. Thus the overall oracle ratio 0.923077 is not a quota-invariant effect.

MovieLens-1M contributes two deliberately limited forms of non-synthetic structure. First, its 1000209 observed user–movie interactions form a connected support with 6,040 users and 3,706 rated movies. All 5 planted numerical stress instances on that support certify, closing the synthetic-topology gap but making no recommender effect claim.

Second, the user–genre panel retrospectively retains the common support having observations on both sides of each cutoff, then derives positive two-

Table 3 Conditional MSE divided by quota-matched uniform MSE at materialized binary64 probabilities. The all-cells column is the fixed-grid geometric mean.

Policy	All cells	Quota ratio 1	Quota ratio 10
Current-norm oracle	0.923077	0.996607	0.854972
Historical norm, age 1	0.923219	0.996767	0.855097
Historical norm, age 20	0.929779	0.998781	0.865543
Coordinate proxy	1.8018	1.97158	1.64665

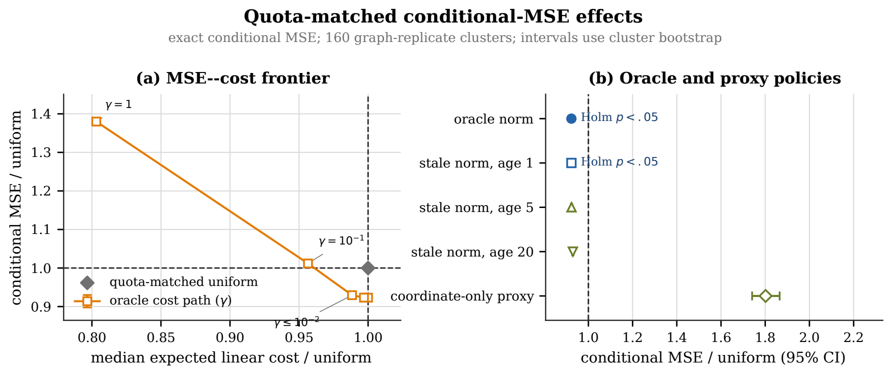

Fig. 7 Synthetic quota-matched conditional-MSE ratios. Confidence intervals cluster by independent graph/update replicate; the dashed line is uniform sampling.

coordinate update proxies from historical and later temporal rating summaries. Five temporal cutoffs, two expected-quota regimes, and three policies give 30 strict endpoints and 10 paired cells. The current-weight oracle has overall conditional-MSE ratio 0.924997 against equal edge weights; the historical proxy has ratio 1.00034 against equal and 1.08145 against the current oracle. Under equal model quotas, the current/equal and historical/equal ratios are 0.918347 and 0.973635; under popularity-aware quotas they are 0.931694 and 1.02778. The historical proxy therefore helps relative to equal weights only in the equal-quota regime, while current weights help in both (figure 8).

The two case studies establish that certified reciprocal allocations can be materialized and evaluated on both controlled update fields and a public sparse interaction structure. They do not establish federated-learning accuracy, training-time, or recommender-system state of the art.

8.5 Tree sensitivity and a historical-only public workload

The disjoint A43 amendment first holds each candidate fixed and changes only the spanning tree. Its 36 graph–candidate pairs cross three graph families, two scales, three seeds, and two registered perturbation radii; four tolerances and four preregistered tree rules produce 576 strict replays. Every replay proves positivity, so tree choice does not alter the soundness theorem. It does materially

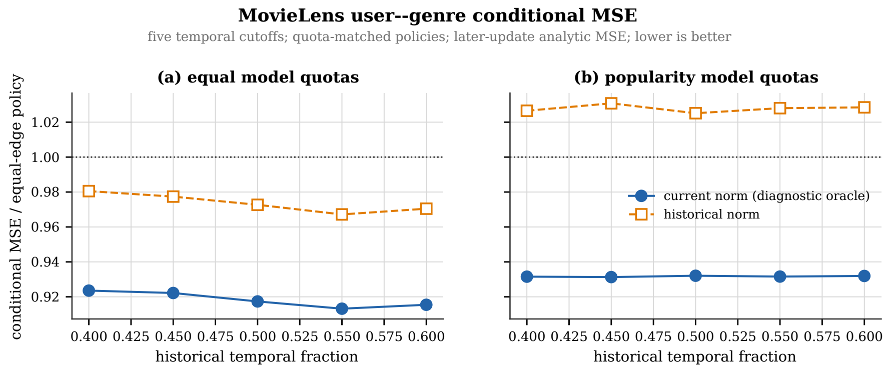

Fig. 8 MovieLens user–genre conditional-MSE ratios over five temporal cutoffs on retrospectively selected common support. This is a descriptive allocation study: it does not train a neural model or evaluate recommendation accuracy.

Table 4 A43 spanning-tree sensitivity on identical candidates. Positivity is certified in all 144 replays for every rule; acceptance additionally requires the registered objective tolerance. The last column is the median maximum tree-edge correction divided by the uncorrected allocation.

Tree rule	Accepted	Positive	Median relative correction
Input order	35/144	144/144	0.01055
Maximum conductance	126/144	144/144	0.002504
Minimum conductance	22/144	144/144	0.01859
Seeded random	36/144	144/144	0.006360

alter sharpness: 104 of 144 candidate–tolerance cells have different acceptance decisions across trees. The median and 95th-percentile within-cell ratios of largest to smallest finite gap bound are 2.06×10^7 and 2.14×10^8. The registered tree-insensitivity gate therefore fails by a wide margin.

Maximum-conductance Kruskal is never rejected in a cell where another rule is accepted. It accepts all 36 candidates at each of relative tolerances 10^{-10}, 10^{-8}, and 10^{-6}, and 18 of 36 at 10^{-12}, for 126/144 overall. This is an empirical selection rule, not a theorem that it minimizes the certificate bound. For new workloads we therefore recommend selecting a maximum-conductance tree from the candidate state, then compiling and freezing that tree before endpoint replay. Its construction cost is outside the fixed-topology linear-work claim and must be reported separately.

The second A43 panel removes future-dependent support selection. At each cutoff it reads only historical MovieLens rows, retains user–genre pairs with at least two historical ratings in the largest connected component, derives μ from two historical rating coordinates, and induces both marginal families from historical pair counts. It starts from the public `dual_feasible_start`; no planted optimum or future potential enters the problem. All 3 workloads receive an endpoint certificate on their first damped-Newton attempt (table 5).

Table 5 A43 MovieLens workload using ratings at or before each cutoff only. All instances have 18 active genres. Times are single registered-machine observations and are descriptive, not a scaling comparison.

Cutoff	Users	Edges	Gap upper bound	ℓ_1 residual	Seconds
0.40	2,664	35,959	7.87×10^{-17}	3.00×10^{-14}	0.154
0.50	3,254	44,129	1.02×10^{-21}	6.25×10^{-14}	0.202
0.60	4,081	54,567	2.23×10^{-22}	4.51×10^{-14}	0.279

The first accepted result is also exported as exact binary64 arrays, candidate, tree, tolerance, and a canonical SHA-256 payload. A fresh reconstruction reruns the strict backend, accepts, and exactly matches the serialized claim fingerprints. This closes the process-local usability gap without turning the JSON summary into authority. The panel establishes a public, historical-input optimization workload and portable endpoint replay; it does not evaluate recommendation accuracy or causal deployment performance.

8.6 Experimental external verified-conic baseline

The disjoint A44 amendment executes VSDP 2020.1 as the closest identified general verified-conic comparator on small and medium instances. Each reciprocal term is represented by one three-dimensional second-order cone and five lifted variables; power-of-two scaling keeps the transformation exact for the represented binary64 coefficients.

The VSDP and SeDuMi sources are pinned upstream releases. A thin adapter maps VSDP's SOCP uses of INTLAB onto the GNU Octave interval package. This is an experimental open-backend port, not INTLAB and not an official stock-VSDP configuration. Before the confirmatory run, it passed a sub-ulp outward interval-construction test, a verified linear-solve containment test, and the VSDP authors' bundled SOCP rigorous-bound assertions.

All 18 preregistered cells cross three graph families, two edge counts, and three seeds. Their complete represented-problem fingerprints match the corresponding immutable A42 CertQuota cells. The experimental port reported finite lower and upper objective bounds in 18/18 cells. Every bracket contained the floating planted reference objective, used only by the analysis. This reference is not an exact optimality witness for the represented input: rounding the constructed flow, costs, and marginals generally breaks exact KKT equalities. Reference containment is consequently a consistency diagnostic, not a correctness test. Moreover, all 18 A44 interval widths exceed the paired A42 requested tolerances (by factors 10.23–121.03); finite-bracket success must not be read as matched-accuracy success. Table 6 reports interval width and resource use. The two methods have different process and runtime boundaries, so this amendment does not authorize a direct speed ratio.

Thus A44 supplies an executed general conic rigorous-bound path and removes the earlier absence of any such run at the algorithm level. It does not

Table 6 A44 experimental VSDP 2020.1/GNU Octave interval-port results on the 18 source-bound confirmatory cells. Each size contains three graph families and three seeds. Time is a fresh Octave process per cell and RSS is the observed process-tree peak; neither is directly comparable with the historical in-process A42 timings.

Edges	Rigorous brackets	Median relative width	Median seconds	Median MiB
300	9/9	2.02×10^{-8}	4.98	219
1000	9/9	7.07×10^{-8}	29.45	1705

close the narrower stock VSDP+INTLAB comparison: a licensed INTLAB installation was not available, and the experimental rows remain ineligible for an official or direct speed-ranking claim.

8.7 Independently witnessed, matched-accuracy revision

A45 is an exploratory response to the preceding limitations, not a new preregistered confirmation of the already inspected A44 instances. A disjoint 120-edge pilot (seed 45001, three families) tests SeDuMi tolerances $10^{-10}, 10^{-12}, 10^{-14}$ and retains all nine attempts. Matched, independently proved brackets occur in respectively 0/3, 3/3 and 2/3 cases, selecting 10^{-12} before the 18-instance rerun. The new runs use exactly the paired A42 problem fingerprints and requested ε values, not an optimizer status or a floating planted objective as ground truth.

Independent witnesses. A standard-library-only checker uses integer-directed dyadic arithmetic at 160 fractional bits. For endpoint bundles it directly encloses $P(Z)$ and $D(y)$, without the production interval backend or its cancellation identity. For each experimental VSDP-port bracket, it reconstructs the lifted dual inequalities from the original data, verifies the LP and squared Lorentz-cone conditions in exact rational arithmetic, and checks the correctly rescaled conic dual objective against L. Separately, exact rational leaf elimination of an exported primal midpoint supplies a positive feasible original flow whose outward objective bound is at most U. Thus the saved witnesses prove that particular original-input bracket even without trusting the port's interval linear solve. They do not validate every operation of the compatibility layer or establish a stock INTLAB result. Additional tests use ten genuinely exact dyadic KKT fixtures and 252 exact right-hand-side corner solves across 12 small linear systems; floating planted reference containment is never used as a correctness gate.

Common accuracy and cost boundaries. Both pipelines run in fresh single-thread processes under 300-second and 4096-MiB sampled-RSS limits. Common MAT export is excluded; the port's external dyadic lift construction is charged separately. Verification time includes VSDP's internal perturbed re-solves, or CertQuota's cold tree/plan preparation and replay. The transfer test uses the

Table 7 A45 first-setting comparison; times are medians over all nine cells at each size, including accuracy failures. Port denotes only the experimental VSDP/Octave-interval implementation, whose linear-solve path densifies. CQ-transfer certifies all 18 initial SeDuMi duals; the full CQ column uses its own public cold candidate pipeline. These are not stock-VSDP speed estimates.

m	Matched / 9		Verification (s)		Audited pipeline (s)	
	Port	CQ	Port	CQ-transfer	Port	CQ
300	9	9	3.506	0.008	6.085	0.827
1000	6	9	31.265	0.012	34.440	1.019

same initial SeDuMi dual, without a CertQuota repair. Full pipeline time adds process startup and an independent witness-check process; these audited pipelines are distinct from warm replay throughput. Peak RSS below concerns the numerical worker, not the separate integer audit. Timings describe these implementations on the recorded host.

All 18 port brackets have independently checked witnesses; 15/18 meet the common target, whereas both CertQuota's full pipeline and the unrepaired SeDuMi-dual transfer meet it in 18/18 cases. The remaining three port widths are 1.77–2.22 times their targets. A fixed second setting, 10^{-14}, is tried for every first-setting failure, charging both attempts to the same time budget; it adds no matched success. All attempts are retained. This is not a lower bound on the accuracy obtainable by VSDP after other tuning. At 1000 edges, median worker RSS is 1728.133 MiB for the port and 68.984 MiB for CertQuota; the dense compatibility path prevents attributing that difference to stock VSDP. The defensible computational conclusion is narrower: on these identical inputs, an original-array endpoint check accepts the same approximate duals with substantially less measured verification work than this executed conic path, at the stated common guarantee.

8.8 A historical decision workload with an explicit rational output

The revision uses three additional MovieLens history cutoffs $(0.45, 0.55, 0.65)$. Every support edge, reciprocal weight and processing proxy uses only pre-cutoff records. Client budgets are fixed to the represented value 0.2. Genre quotas are specified before optimization as $(1-\theta)d^{\text{hist}} + \theta(0.2n_{\text{left}}/n_{\text{genre}})\mathbf{1}$, with $\theta \in \{0, 0.25\}$ and the same exact reduced-root convention. Historical rating counts k_e define nonzero costs $c_e = \lambda k_e$; λ is normalized so that baseline processing cost is approximately 0.1 or 1 times baseline reciprocal cost. These are explicit task-design proxies, not measured production latencies. Raking supplies an approximate baseline, which is repaired and checked as an exactly feasible rational flow before it is used for the policy comparison.

Let $[L_0, U_0]$ enclose that feasible baseline's objective. The pre-specified decision asks whether a feasible allocation can meet the cap $0.99L_0$; the requested gap is $10^{-8}(1 + U_0)$. A rational allocation $\widehat{Z}$ is exported with all numerators and denominators, rather than claimed to be bitwise feasible in binary64. Its independent checker tests $B\widehat{Z} = \beta$ exactly, strict positivity, and an objective interval. It returns "below cap", "cap infeasible", or "undecided" according to the proved upper and lower endpoints, never a rounded point estimate. All 12/12 combinations certify, and all prove at least the requested 1% improvement over their own feasible baseline. The conservative improvement bounds range approximately from 41.5% to 53.5%. This demonstrates a usable mathematical policy decision and a transferable exact allocation; it does not establish a statistical improvement in a deployed recommender or MMFL training system. The 12 combinations share one dataset, not 12 independent applications.

8.9 Hard starts and dynamic tree rebuilding

Nine new family–seed graphs (seeds 45101–45103) discard the generator's KKT-planted costs and replace them by independently drawn positive costs. Two reciprocal-weight spans, quota shocks, public cold starts and candidates within 10^{-10} relative slack of the dual-domain boundary give 36 valid hard-start cases (18 represented problems, two starts per problem). CertQuota independently certifies 16/18 public cold starts and 10/18 near-boundary starts under a 20-second budget. All 18 deliberately out-of-domain controls are rejected. Failed candidates and repairs are retained separately from accepted endpoints; these tests do not replace the earlier warm-start scalability study. All ten valid failures receive an explicit optional follow-up: a CVXPY/Clarabel candidate with a 10-second budget followed by up to 10 seconds of the unchanged strict pipeline. Seven certify. The remaining three conic duals, despite an approximate optimal status, lie outside the original dual domain; convex retraction toward a public domain point and another bounded refinement certify all three. Thus 36/36 valid starting cases eventually have independently checked endpoints, while the original 26/36 result and all additional work remain visible. This is an exploratory repair study with extra budget, not a claim that the unchanged 20-second default succeeds universally.

The same held-out family–seed grid then exercises eight changing rounds and three tolerances. Three explicit policies keep the initial maximum-conductance tree, rebuild it every round, or rebuild only after a failed replay. Each policy has 216 candidate–tolerance cells. Construction, cold router plans, preparation and retries are charged, policy order rotates, and candidate generation is recorded separately. Multiple rounds and tolerances are not independent graph samples. The production default was already maximum conductance before this revision; the new evidence concerns when to reuse or rebuild, not the discovery of that default. Rejection-triggered rebuilding remains a measured policy with an explicit extra cost, not part of the fixed-topology complexity theorem. On this grid it matches the always-rebuild acceptance count using 66 rather than 216 builds, with 1.112 rather than 1.943 seconds total policy cost. The frozen policy is cheaper but rejects 38 cells. This makes the reuse–rebuild tradeoff explicit; no tree is asserted to be uniformly best.

Table 8 Held-out A45 dynamic-policy comparison. Times include unsuccessful checks and rebuilds but exclude the common candidate generator. These are finite policy observations, not a universal optimal-tree guarantee.

Policy	Accepted / 216	Tree builds	Total policy time (s)
Frozen tree	178	27	0.800
Rebuild every round	216	216	1.943
Rebuild after rejection	216	66	1.112

9 Related work

Reciprocal transport and convex network flow. The objective itself is not new. The β-potential formula of Dessein et al. [16], extended algebraically beyond its analyzed parameter range, gives

$$\phi_{-1}(z) = \frac{z^{-1} + z - 2}{2}, \qquad c_e z_e + \frac{\mu_e}{z_e} = (c_e - \mu_e) z_e + 2\mu_e \phi_{-1}(z_e) + 2\mu_e.$$

Thus our weighted objective is a formal edge-weighted $\beta = -1$ specialization after an edgewise shift of the linear costs and an additive constant. This is an algebraic identification, not an inherited algorithmic guarantee: Dessein et al. [16] analyze $0 < \beta < 1$ for this family and explicitly note that other parameter values need not satisfy their assumptions. Newton and interior methods for separable nonlinear network and transportation problems are classical [26, 34, 37, 38]. Earlier primal algorithms already maintain feasible convex-cost flows and derive objective upper bounds: the method in [57] uses the most negative incremental-cost circuit both to update a feasible flow and to bound its remaining suboptimality. The ϵ-relaxation method of [7] is globally terminating and near-optimal with a complexity analysis. These are solver algorithms with their own iterates and termination logic, rather than posterior outward-rounded replay contracts for arbitrary candidate arrays.

Primal recovery from a nonoptimal dual vector is also not new: [41] constructs primal-feasible flows for separable strictly convex single-commodity networks using residual-graph and shortest-path schemes, with convergence to the unique primal optimum along a convergent dual sequence and an associated primal–dual stopping criterion. Their construction is therefore also usable around dual algorithms rather than tied to one candidate generator. Its real-arithmetic recovery is not an outward-rounded finite endpoint replay, does not require edgewise strict positivity, and does not state an original-array $O(m + n)$ certificate bound. More recent work exploits bipartite normal equations, sparsity, and regularization at far greater generality [11, 15, 54, 56, 58]; almost-linear flow algorithms address broader objectives [13]. Finite-precision primal recovery and explicit gaps also appear in an almost-linear solver for a different hypergraph-Laplacian energy [60]. We do not claim a new Newton method, Laplacian Hessian, finite-precision graph recovery, reciprocal formulation, or globally near-linear candidate solver. The Convex Network Flows framework covers a broader class of hypergraph flow models and supplies an edge-decomposed algorithm and the open-source ConvexFlows.jl implementation [17]. It is therefore a relevant general solver and software precedent, but does not make the signed original-array endpoint claim studied here. Recent OT candidate solvers combine truncated or sparse Newton steps with Sinkhorn or Bregman proximal frameworks [36, 47, 59]; a contemporaneous network-equilibrium preprint reports empirical near-linear scaling for a different undirected model [44]. These works strengthen candidate generation and global solution, whereas our theorem concerns a finite, solver-independent endpoint replay on the represented reciprocal problem. A recent dynamic-OT Newton method uses a spanning-tree gauge to remove cycle redundancy and a separate CFL positivity condition [14]; that use of a tree is distinct from our support-preserving endpoint residual repair and precludes any claim that trees or positivity are new to OT Newton methods.

Approximate transport algorithms often postprocess marginals. The rounding construction in [2], for example, can add a residual outer product and need not preserve a prescribed sparse eligibility support. Our spanning-tree correction instead stays on declared edges and couples exact marginal repair to an explicit strict-positivity and objective-gap test. That support-preserving certificate, rather than rounding in general, is the structure used here.

Verified optimization. Separating a fast computation from a checkable witness is the certifying algorithm paradigm [45]. Rigorous objective bounds and postprocessing exist for convex, conic, linear, and semidefinite programs; general accuracy certificates formalize how computed information can authorize quantitative error statements [28–31, 40, 46, 51]. Interval methods can separately prove equality-constrained feasibility [35], while modern certificate formats and formally verified floating-point kernels address proof portability and trusted implementation boundaries in other problem classes [39, 53]. General conic dual certificates can also reconstruct or certify primal cone membership, and feasible nonsymmetric conic interior-point methods can produce compatible near-optimal candidates [42, 48]; therefore neither posterior certification nor dual-to-primal reconstruction is new in principle. Fast feasible primal recovery from dual iterates is also studied beyond network flow [49], and accelerated primal–dual averaging can produce general convex accuracy certificates [10]. Verified DCP/conic reductions in Lean address the complementary modeling boundary [6]; our artifact does not claim a formally verified reduction from an application model. Standard conic quadratic formulations and primal–dual implementations [3, 19, 23] apply to equation (1) after introducing one cone block per edge.

VSDP is the closest identified general rigorous software comparator: it can verify linear, second-order-cone, and semidefinite programs, but its documented runtime requires MATLAB or GNU Octave, INTLAB, and an approximate conic solver [32]. None of MATLAB, Octave, or INTLAB was present in the frozen A42/A43 environment, so stock VSDP was unavailable there rather than encoded as a synthetic timeout. A44 subsequently pins VSDP and SeDuMi and executes the SOCP rigorous-bound routines through an audited experimental

GNU Octave interval adapter. The adapter result is reported explicitly as an open-backend port and is not relabeled as stock VSDP+INTLAB. The executed CVXPY/Clarabel lift measures candidate transfer, while Arb measures an independent replay of our predicate; neither is a general verified optimizer. A44 removes the lack of an executed generic conic bound computation. A45 adds exact per-instance witness checks, common tolerances and fresh-process timing. The absent official INTLAB configuration and the port's dense interval path still prevent a general performance claim against verified conic software.

Our narrower distinction is the simultaneous contract of an endpoint replay on the original (B, β, c, μ) arrays: no m-block conic lift or generator status is trusted, tree recovery preserves sparse support, signed outward intervals prove strict positivity and a gap for the same implicit exact-real allocation, and the fixed-topology replay has $O(m+n)$ work and memory. Feasible primal recovery, objective-gap bounds, interval verification, trees, and fast convex-flow software all have prior art; the contribution is their structure-specific combination in one finite endpoint decision procedure. Arb-V is a separately implemented higher-precision audit of the same endpoint predicate [33]; it is neither an external method nor evidence that general verified conic software is inferior.

Table 9 separates capabilities already present in prior work from their combination in our endpoint predicate. Entries marked *no endpoint claim* are outside the cited method's stated output contract; they do not assert impossibility within that broader framework.

Federated client sampling. Concurrent multi-model FL allocates limited clients among models [4, 8, 9]. The variance-aware MMFL sampling line includes an ICDCS poster and its expanded MMFL-LVR/StaleVR formulation [61, 62]. It uses client capacities and a global participation total, which permits a different closed-form allocation; it does not impose one participation marginal for every model. Fair concurrent training controls model-level allocation [52], while group, clustered, adaptive, and engagement-aware methods address other statistical or systems objectives [20, 22, 43, 63]. Market-equilibrium work allocates fractional client resources across multiple server models with fairness and efficiency guarantees [18], but optimizes utilities rather than inverse-probability variance under fixed model marginals. Our reduction occupies their intersection only under continuous two-sided *expected* quotas and independent categorical client decisions.

Balanced survey sampling and dependent rounding study related inclusion probabilities and realized constraints [12, 21, 25, 50, 55]. They also make clear why exact integer model counts in each round would change the covariance structure and require a separate analysis.

Table 9 Relation to the closest method families. “No finite endpoint” means that the cited work does not state the simultaneous outward-rounded guarantee in the third column; it is not an impossibility claim.

Method family	Native object	Finite endpoint contract	Distinction here
Reciprocal transport and classical convex network flow [7, 16, 37, 38, 41, 57]	Transport or network arcs; support is native	Feasible iterates, recovery, and stopping bounds in real arithmetic; no single outward-rounded arbitrary-candidate endpoint	Same sparse support, but one finite replay binds positivity, exact-real feasibility, and the gap
General convex-flow solvers [13, 17]	Graph or hypergraph flow model	Solver-dependent global guarantees; no arbitrary-candidate endpoint replay	Original reciprocal arrays and generator-independent return authority
Transport rounding [2]	Dense transport plan	Exact-arithmetic marginal repair; strict positivity and endpoint gap are not the target	Tree repair remains on the declared sparse support
Verified conic and interval optimization [28–32, 40, 51]	Generic lifted cone or nonlinear/KKT enclosure	Rigorous formulation-dependent bounds; potentially broader	No trusted m-block lift; fixed-topology replay is $O(m+n)$
General accuracy certificates and conic recovery [10, 42, 46, 48, 49]	Generic convex information or cone representation	Quantitative certificates or recovery under method/model assumptions	One structure-specific original-array decision procedure
This work	Sparse (B, β, c, μ) arrays and a spanning tree	Mandatory outward-rounded positivity, exact-real feasibility, and ε-gap for one implicit allocation	Self-contained replay input, support preservation, and linear fixed-topology work/memory

10 Limitations

The theorem assumes a connected declared graph, finite inputs, and $\mu_e > 0$ on every declared edge. Global candidate routines additionally target instances with bipartite-sign marginals and positive feasibility; endpoint acceptance itself supplies that feasibility for every authorized represented instance. Removing a zero-weight eligibility relation changes the problem support; flooring it changes the objective. The core routine rejects malformed inputs but is not a general sparse-transport feasibility oracle.

Endpoint soundness is conditional on theorem C.1: correctly rounded binary64 primitives and square root, gradual underflow, and the fixed expression DAG. Runtime probes are health checks, not a proof that every future execution satisfies those assumptions. The implementation fails closed, so a mathematically valid candidate can be rejected at fixed precision. The theorem authorizes an implicit exact-real allocation; a materialized binary64 vector has a measured, not bitwise-zero, residual. Likewise, an MMFL sampler must realize the certified inclusion probabilities rather than silently renormalize them. The A45 companion additionally exports explicit rational allocations and checks their exact marginals and objective bounds. This removes the implicit-object ambiguity for those exported workloads, at an extra arbitrary-precision cost; it does not make the original binary64 vector exactly feasible or supply an exact categorical-sampling implementation. Candidate-generator independence does not provide process isolation: arbitrary finite numeric output is allowed, but concurrent mutation of replay snapshots, private prepared-state capabilities, verifier code, or floating-point controls is outside the contract. Public arrays are bytes-backed, public boundaries recheck object seals, and malformed or stale capabilities fail closed under the ordinary API; hostile Python reflection is not a security claim.

The conditional $\Omega(m)$ result concerns accepting probes of fresh explicit slack-sensitive edge arrays. It excludes authenticated preprocessing, proof oracles, and always-reject programs, and says nothing about the total cost or global convergence of a candidate solver. The dynamic experiments use supplied smooth and shock warm starts on one workstation. The full-pipeline scaling panel likewise starts from a planted dual with a 5% relative-slack perturbation, fixes quota ratio one, zero linear cost, and reciprocal-weight span four, and excludes cold graph/router/topology-plan construction. It is therefore a warm-start public-call scaling study, not evidence for arbitrary initialization or first-use global solver complexity. The internal strict methods and conic references are a limited named set; the independent Arb checker replays the endpoint decision but is not a general verified conic optimizer. Candidate rejection can reflect either candidate quality or finite-precision conservatism. Certificate soundness holds for any validated spanning tree, but acceptance sharpness, correction magnitude, and timing can depend on the tree; the A43 panel confirms substantial dependence, with decision differences in 104 of 144 candidate–tolerance cells. Maximum-conductance Kruskal dominates the other registered rules on acceptance in this grid, but no theorem makes it optimal and other graphs may favor another tree. Tree construction and any search over trees are outside the fixed-topology replay bound. Together with the absence of a stock VSDP+INTLAB run, these facts preclude a ranking over all transportation or verified-optimization software.

The MovieLens user–movie panel uses a real eligibility topology but planted optimization quantities and therefore does not evaluate a recommender. The user–genre panel derives update proxies from rating histories, but its endpoint is still conditional sampling MSE rather than held-out ranking or trained-model performance. Its eligibility support is selected retrospectively by requiring observations both before and after each temporal cutoff; later data therefore participate in support construction. This preserves the reported arithmetic comparison on that post-selected support, but it is not a leakage-free deployment simulation. A43 separately removes that leakage for the optimization input: support, weights, quotas, start, and tree use pre-cutoff ratings only, and all three instances certify. Its genre quotas are induced from historical counts to guarantee feasibility, its linear costs are zero, and it still measures only an optimization workload, not held-out recommendation quality. A45 adds nonzero processing-cost proxies, externally specified coverage mixtures, and a certified improvement-over-policy decision. Its 12 cells still reuse one dataset and use designed costs and quotas, not measured production latency or independent application deployments. In the earlier conditional-MSE study, current-norm weights require unavailable future information and serve only as a diagnostic oracle; historical and coordinate weights optimize surrogates. Quotas hold in expectation, not as exact per-round integers. No experiment measures neural-network accuracy, demographic fairness, communication-to-target, or end-to-end MMFL training time. Consequently, the paper makes neither a general optimization SOTA claim nor an MMFL training SOTA claim.

The historical timing evaluation is tied to the declared Windows/AMD environment. A later public-release check builds and installs a non-editable wheel on Ubuntu 24.04 and Windows 2022, with 35 core tests and three independently checked 1024-edge examples on each platform. This is an executed portability check, not a complete cross-platform rerun of the historical experiments. No macOS run or universal hardware guarantee is claimed.

11 Software and artifact availability

The accompanying `certquota` 0.2.0 source is an installable Python package under the BSD-3-Clause license. Its stable public namespace exposes the problem type, verified options and result records, `solve_verified`, and the portable replay writer/checker. A standard `pip install .` installs from the supplied source tree; the release audit also builds the source distribution and wheel and applies package metadata checks. The authoritative call returns an immutable `VerifiedResult`. Only `status == "optimal"` carries a strict certificate, its bound problem/router snapshot, and a materialized allocation. Invalid domains, timeouts, generator exceptions, and exhausted repair cascades return explicit nonoptimal statuses with no authorized allocation; malformed constructor or low-level arguments may raise validation exceptions.

The A42 evidence bundle is content-addressed twice: an execution-source manifest binds code and protocols before formal production, and a submission manifest binds the resulting raw analyses, figures, tables, package inputs, and manuscript. The read-only audit described in appendix H rehashes both closures and regenerates their derived products. A43 has a separate frozen source hash (prefix `e78292a27473`), raw/summary hashes, deterministic analysis, and portable replay; it is not pooled into A42. The A44 confirmatory execution has a separate 44-file source manifest (SHA-256 prefix `57512c85b5a0`), exclusive-create raw/availability/summary records, an independent deterministic analysis, and a retained compatibility validation record. Its stock-claim flag remains false. A44 is not pooled into A42 or A43 and does not modify their historical timing contracts. A45 preserves the 127-file A44 historical closure in a hash-checked archive before editing the current manuscript. Its new records, independent integer/rational checkers, explicit rational allocations, and optional conic-candidate portfolio are a separate revision overlay. A clean Windows virtual environment installs the non-editable wheel from original dependency archives checked against their PyPI SHA-256 values, passes 35 installed-package tests without the checkout's path-modifying test configuration, and independently checks three 1024-edge public-cold-start examples. The development suite passes 404 tests with no skips. Public GitHub Actions run `34040622682`, on source commit `dfd58e1d5644`, subsequently passes the isolated installed-wheel checks on both Ubuntu 24.04 and Windows 2022. Each job uses Python 3.11, NumPy 1.26.4, and SciPy 1.15.3, passes the 35-test subset, and independently checks three 1024-edge examples; its logs, exact runtime details, wheel hash, and platform probe are retained with the public release. This closes the clean Linux installation check only, not the full historical-replication scope. Automated numerical validation is not a human proof review. The checksum-constrained MovieLens-1M archive is not redistributed; all MovieLens analyses require a separately obtained matching archive. The public repository and dated preprint release are listed in the declarations. Its curated publication snapshot supplies core installation tests and standalone proof verification, but is not a byte-for-byte copy of every historical local execution environment. Historical audit records describe their own frozen snapshots, not this later author-identified preprint. No archival DOI is claimed.

12 Conclusion

Sparse reciprocal transportation permits a small, explicit boundary between untrusted numerical work and strict return authority. A finite candidate potential, a support-preserving tree correction, and a signed outward-rounded replay on the original arrays jointly certify positivity, exact feasibility for the represented marginals, and an objective-gap bound for one implicit exact-real allocation. On a fixed topology, the endpoint replay requires linear work and memory, including under adversarially skewed reduction groups. These guarantees depend on the declared arithmetic and ownership contract, but not on a candidate solver's status or stopping logic.

CertQuota-V turns that theorem into a fail-closed software interface: fast generators propose candidates, while a fresh terminal replay alone authorizes a return. The A42 study separates predicate sharpness, structural complexity, complete-pipeline cost, generator quality, and sampling utility, rather than collapsing them into one solver-ranking claim. Its content-addressed evidence includes adversarial topology tests, dynamic and scaling panels, independent high-precision replays, and synthetic and MovieLens conditional-MSE studies. The A43 amendment also exposes a material tree-selection effect instead of hiding it, identifies maximum-conductance Kruskal as the strongest registered rule on its grid, and certifies three prospectively constructed MovieLens workloads with a third-party replay bundle. A44 additionally executes an experimental open-backend VSDP rigorous-conic baseline on 18 source-bound instances, while preserving the distinction from stock VSDP+INTLAB and declining a cross-runtime speed ranking. A45 strengthens the computational case through independently witnessed matched-accuracy comparisons, explicit rational policy decisions, and charged dynamic-tree policies. Difficult starts expose failures of the original candidate cascade; documented optional conic retries and domain retraction recover these cases without changing the endpoint acceptance rule. The resulting contribution is a reproducible verified numerical primitive for a specific sparse convex problem. It does not establish a globally near-linear candidate solver, universal runtime superiority, or state-of-the-art federated-learning training.

A MMFL variance and descent calculations

For fixed model s, write $v_{is} = \alpha_{is}u_{is}$ and $X_{is} = Z_{is}/p_{is} - 1$. Then $\widehat{U}_s - U_s = \sum_i X_{is}v_{is}$. All expectations in this section are conditional on $\mathcal{A}_t$ from section 2. Thus $\mathbb{E}[X_{is} \mid \mathcal{A}_t] = 0$, different clients decide independently, and cross terms vanish:

$$\mathbb{E}\left\|\widehat{U}_s - U_s\right\|_2^2 = \sum_i \mathbb{E}[X_{is}^2]\, \|v_{is}\|_2^2 \tag{18}$$

$$= \sum_i \left(\frac{1}{p_{is}} - 1\right) \alpha_{is}^2 \|u_{is}\|_2^2 . \tag{19}$$

This proves theorem 2.1. Within a client, indicators for different models are negatively related, but no cross-model covariance enters a sum of separate per-model squared errors. A joint norm mixing models would require its own covariance calculation.

This is a potential-update conditional identity. If local updates retain randomness after the scheduling decision, the same formula requires conditional independence between selection and the potential update, with $\|u_{is}\|_2^2$ replaced by its appropriate conditional second moment. Without that condition the reciprocal variance objective need not be exact.

For completeness, if f_s is L_s-smooth and $U_s = \nabla f_s(w_s)$, then for any step $\gamma_s > 0$,

$$\mathbb{E} f_s(w_s - \gamma_s\widehat{U}_s) \le f_s(w_s) - \gamma_s \|U_s\|^2 + \frac{L_s\gamma_s^2}{2}\mathbb{E}\left\|\widehat{U}_s\right\|^2 \tag{20}$$

$$= f_s(w_s) - \gamma_s\left(1 - \frac{L_s\gamma_s}{2}\right)\|U_s\|^2 + \frac{L_s\gamma_s^2}{2}\mathbb{E}\left\|\widehat{U}_s - U_s\right\|^2 . \tag{21}$$

Thus variance optimization minimizes the sampling-dependent term in this one-step upper bound when quotas and step sizes are fixed. It does not by itself prove an end-to-end training advantage.

B Well-posedness and dual derivation

Let the left and right marginals be strictly positive, let every c_e be finite, let every μ_e be finite and strictly positive, and suppose there exists a strictly positive feasible allocation. The closure of the bipartite transport polytope is compact because $0 \le z_{is} \le \min\{r_i, d_s\}$. Since $\mu_e/z_e \to +\infty$ at the boundary, each finite objective sublevel stays in the positive interior. Strict convexity,

$$\nabla^2 P(z) = \operatorname{diag}(2\mu_e/z_e^3) \succ 0, \tag{22}$$

then gives a unique primal optimum.

Delete the distinguished root row of B and the corresponding component of β. Connectedness makes the resulting matrix $\widetilde{B}$ full row rank. Thus the reduced affine equality system satisfies the linear-independence constraint qualification everywhere. The unique optimum lies in the positive interior, so the equality-constrained KKT conditions provide a multiplier $\widetilde{y}^\star$. After restoring an arbitrary root gauge, stationarity is

$$c - B^\top y^\star = \mu/(z^\star)^2 > 0. \tag{23}$$

Consequently the dual domain is nonempty, the coordinatewise Lagrangian infimum is attained at $z^\star$, and primal and dual objective values agree. In particular, strong duality and dual attainment used below follow from the affine constraint qualification rather than from a numerical solver status.

Under the sign convention in the main text, the Lagrangian is

$$\mathcal{L}(z,y) = c^\top z + \sum_e \frac{\mu_e}{z_e} + y^\top(\beta - Bz). \tag{24}$$

For $q_e = c_e - (B^\top y)_e > 0$, minimizing one coordinate gives $z_e = \sqrt{\mu_e/q_e}$ and value $2\sqrt{\mu_e q_e}$. This yields equation (8). The dual formula and weak duality do not require a prior feasibility oracle. For every y with $q(y) > 0$ and every positive z satisfying $Bz = \beta$,

$$P(z) = \mathcal{L}(z,y) \geq \inf_{u>0} \mathcal{L}(u,y) = D(y).$$

The strictly positive feasibility assumption at the beginning of this section is used only to establish primal attainment, KKT multipliers, and dual attainment. In theorem 4.1, that assumption is discharged after acceptance by the recovered $Z > 0$.

Differentiation gives

$$\nabla D(y) = \beta - Bx(y) = -g(y), \qquad \nabla(-D)(y) = g(y), \tag{25}$$

and

$$\nabla^2(-D)(y) = H(y) = B\,\mathrm{diag}\left(\frac{\sqrt{\mu_e}}{2q_e^{3/2}}\right)B^\top. \tag{26}$$

Because the graph is connected and all conductances are positive, H is positive definite on $\mathbf{1}^\perp$; equivalently, $\nabla^2 D = -H$ is negative definite there. Thus D is strictly concave modulo addition of a constant potential.

The exact reduced-quota convention matters here. For the theorem, let v_0 be the distinguished root and let $\bar{\beta}$ denote the independently supplied binary64 quota vector. The represented problem keeps $\bar{\beta}_v$ for $v \neq v_0$ and defines

$$\beta_{v_0} := -\sum_{v \neq v_0} \bar{\beta}_v, \qquad \Delta_{\mathrm{root}} := \beta_{v_0} - \bar{\beta}_{v_0}. \tag{27}$$

The sum in equation (27) is an exact-real definition; the stored binary64 root is only a numerical representative. The implementation chooses $v_0 = n - 1$, rejects nonfinite data and a declared imbalance exceeding $10^{-9}\max\{1, \|\bar{\beta}\|_1\}$, and records the supplied root, an `fsum`-based binary64 representative of the implied root, and the rounded diagnostic $\widehat{\Delta}_{\mathrm{root}}$ with its relative magnitude. The exact-real Δ_{root} in equation (27) is reconstructed from the independent entries by the strict replay and is not a stored floating diagnostic. The constructor does not run a generic transportation-feasibility oracle. With incidence columns positive on the client side and negative on the model side, global candidate routines are designed for bipartite-sign marginals and a positive feasible allocation. Generators begin from a positive binary64 planted flow, but its rounded marginals are not asserted to equal Bx in exact-real arithmetic. For every instance on which an optimality claim is made, endpoint acceptance itself supplies a strictly positive exact-real feasible Z and therefore discharges nonemptiness for that represented instance. A violation can produce only rejection or explicit failure, never a strict accepted allocation. All feasibility and objective claims concern β from equation (27), not the independently rounded vector $\bar{\beta}$. Operationally, the strict replay need not materialize the generally non-binary64 root quota: it computes every non-root residual interval and defines the root residual interval as the directed negative sum of those intervals. This is algebraically equivalent to the exact-real root convention. Consequently $\mathbf{1}^\top g = 0$ algebraically, the tree correction exists, and $Bz = \beta$ means exact feasibility for the represented input.

C Expanded endpoint soundness proof

C.1 IEEE acceptance predicate and inclusion lemma

Assumption C.1 (Execution arithmetic contract) *Every stored scalar is interpreted as its exact finite binary64 dyadic value. Every executed binary64 $+, -, \times, /$ operation and square root used by the authoritative replay is correctly rounded in round-to-nearest, ties-to-even mode; gradual underflow is enabled; and* `nextafter` *returns the adjacent binary64 value in the requested direction. Exact sign changes, absolute values, comparisons, selections, and index operations have their stated scalar semantics. The implementation executes the declared reduction DAG without an unrecorded contraction or reassociation that changes a DAG node. Reduction schedules are compiled from, and remain content-bound to, the same immutable problem and tree snapshots from compilation through replay. Prepared states are accepted only when created by the authoritative in-process factory and their identity/content seals remain valid. Candidate independence permits arbitrary finite numeric output, not mutation of replay memory, code, private capabilities, or floating-point controls during the authoritative call.*

All executable soundness conclusions in this appendix and theorem 4.1 are conditional on theorem C.1. The platform probes are fail-closed health checks: passing finitely many probes does not logically establish the assumption for all operands, threads, compiler paths, or future executions. The requested tolerance is also part of the represented input and must satisfy $0 < \varepsilon < +\infty$.

Let $\mathbb{F}$ be binary64, let fl denote a correctly rounded round-to-nearest operation, and let down and up return the adjacent representable number toward $-\infty$ and $+\infty$, respectively. Degenerate intervals represent stored binary64 inputs as exact dyadics. For $+, -, \times$, and $/$, the verifier uses the standard interval extension, evaluates every required endpoint operation with fl, and pads the lower and upper results by down and up. Division rejects an interval containing zero. For $0 \le L \le U$ it uses

$$\sqrt{[L, U]} := \left[\mathrm{down}(\mathrm{fl}(\sqrt{L})), \mathrm{up}(\mathrm{fl}(\sqrt{U}))\right]. \tag{28}$$

Exact sign changes map $[L, U]$ to $[-U, -L]$. Every grouped sum is a fixed binary tree of these padded interval additions. A NaN, an infinite endpoint, a failed platform-contract check, or an overflow at any node causes rejection.

For completeness, one authoritative replay executes the following fixed DAG.

1. Validate finite c, μ, y, ε, prove $\mu_e > 0$ and $0 < \varepsilon < +\infty$, validate that T spans the graph, and reconstruct the exact-root interval from the independent quota entries according to equation (27).
2. From degenerate input intervals evaluate $Q_e \supseteq c_e - b_e^\top y$. Reject unless every lower endpoint $\underline{Q}_e$ is strictly positive.
3. Evaluate $X_e = \sqrt{[\mu_e, \mu_e]/Q_e} = [\ell_e, u_e] \ni x_e(y)$, then evaluate $G \supseteq Bx(y) - \beta$ with the precompiled client and model reduction DAGs.
4. Apply the fixed tree-routing DAG to G. It produces signed intervals $F_e = [\underline{F}_e, \overline{F}_e]$ enclosing the unique tree-supported flow f_T satisfying $Bf_T = g$. Since $h_T = -f_T$, retain
$$H_e = [\underline{h}_e, \overline{h}_e] := [-\overline{F}_e, -\underline{F}_e] \ni h_{T,e}, \qquad a_e := \max\{|\underline{h}_e|, |\overline{h}_e|\}.$$
Equivalently, a backend may route $-G$ directly. Sign change, absolute value, and selection are exact under theorem C.1. Non-tree correction coordinates are zero.
5. On every tree edge evaluate the directed lower endpoint $\underline{z}_e = \mathrm{down}(\ell_e + \underline{h}_e)$ and reject unless $\underline{z}_e > 0$.
6. Evaluate every nonnegative term
$$\frac{\mu_e a_e^2}{\ell_e^2 \underline{z}_e}$$
outward and sum the terms with the fixed scalar-reduction DAG. Reject unless its upper endpoint Γ_U is finite and at most the requested ε.

7. Accept only after all preceding tests pass. The strict object is the tuple $(P, y, T, \text{certificate})$ at the requested ε; the fixed replay DAG is determined by the validated topology and tree plans. The process-local compact summary binds fingerprints of P, y, T, and the binary64 tolerance, while full edgewise intervals are transient and can be regenerated only by replay.

Lemma C.1 (Primitive and replay inclusion) *Under theorem C.1, on every accepting replay, each stored interval contains the exact-real value of the corresponding node in the replay DAG. In particular,*

$$\begin{aligned}
&Q_e \ni q_e(y), && X_e \ni x_e(y), && G_v \ni g_v(y),\\
&F_e \ni f_{T,e}(y), && Bf_T = g, && H_e \ni h_{T,e} = -f_{T,e},\\
&|h_{T,e}| \le a_e, && x_e + h_{T,e} \ge \underline{z}_e. &&
\end{aligned}$$

Moreover, on acceptance,

$$\Gamma_U \ge \sum_{e\in T} \frac{\mu_e a_e^2}{\ell_e^2 \underline{z}_e} \ge \sum_{e\in T} \frac{\mu_e h_e^2}{x_e^2(x_e + h_e)} = G(y, T). \tag{29}$$

Proof A stored finite binary64 number is an exact dyadic, so the claim holds at every leaf. Correct rounding followed by one predecessor or successor encloses the exact result of a basic operation; the usual interval endpoint formulas preserve inclusion for $+, -, \times$, and for division away from zero. Monotonicity of square root proves equation (28). Induction over each fixed pairwise reduction DAG therefore encloses the exact grouped incidence sums and the exact-root sum. Tree routing contains only additions and sign changes, so the same induction encloses the unique exact tree flow. Absolute-value maxima and the outward arithmetic in the positivity and gap nodes preserve their stated one-sided bounds. The verifier rejects every exceptional nonfinite case, so an accepting trace reaches no operation outside these hypotheses. The outward scalar reduction gives the first inequality in equation (29); $|h_e| \le a_e$, $x_e \ge \ell_e$, and $x_e + h_e \ge \underline{z}_e > 0$ give the second. The final equality is the coordinatewise reciprocal identity expanded below together with $B(x + h_T) = \beta$.

Let b_e denote the oriented incidence column. The strict domain test encloses $q_e = c_e - b_e^\top y$ and proves its lower endpoint positive. Outward square-root evaluation then produces $\ell_e \le x_e = \sqrt{\mu_e/q_e}$. Tree routing is a sequence of exact-real additions and sign changes; applying the same fixed reduction DAG to directed lower and upper endpoints encloses the unique exact tree flow f_T satisfying $Bf_T = g$. Since $h_T = -f_T$, exact sign change gives $H_e = [-\overline{F}_e, -\underline{F}_e] \ni h_{T,e}$ and hence $a_e \ge |h_{T,e}|$.

For accepted tree edges,

$$Z_e = x_e + h_e \ge \ell_e + \underline{h}_e \ge \underline{z}_e > 0. \tag{30}$$

For non-tree edges $h_e = 0$, so $Z_e = x_e > 0$. Since $Bh_T = -g$ by the exact tree equations,

$$BZ = B(x + h_T) = \beta + g - g = \beta. \tag{31}$$

To derive the objective bound without a norm relaxation, use feasibility and $B^\top y = c - q$:

$$\begin{aligned}
P(Z) - D(y) &= c^\top Z + \sum_e \frac{\mu_e}{Z_e} - \beta^\top y - 2\sum_e \sqrt{\mu_e q_e} \qquad (32)\\
&= \sum_e \left(q_e Z_e + \frac{\mu_e}{Z_e} - 2\sqrt{\mu_e q_e} \right). \qquad (33)
\end{aligned}$$

Because $q_e = \mu_e/x_e^2$ and $Z_e = x_e + h_e$, direct algebra gives

$$q_e Z_e + \frac{\mu_e}{Z_e} - 2\sqrt{\mu_e q_e} = \frac{\mu_e h_e^2}{x_e^2(x_e + h_e)}. \tag{34}$$

The term vanishes off the tree. On a tree edge, positivity and the certified one-sided bounds give

$$\frac{\mu_e h_e^2}{x_e^2(x_e+h_e)} \le \frac{\mu_e a_e^2}{\ell_e^2 \underline{z}_e}. \tag{35}$$

Weak duality gives $P^\star \ge D(y)$, and the outward scalar-reduction DAG contains the right-hand sum. This proves theorem 4.1.

Increasing-precision completeness and sharpness. Fix a candidate and tree satisfying $q > 0$ and $x + h_T > 0$ on every edge. For a sequence of interval evaluations of the same finite DAG, write $I_{k,v}$ for the enclosure of exact node value v and assume $v \in I_{k,v}$ with $\operatorname{diam}(I_{k,v}) \to 0$ at every node. This is an interval-continuity hypothesis, not an additional claim about the fixed binary64 implementation. It implies $\ell_{e,k} \to x_e$, both endpoints of $H_{e,k}$ converge to h_e, $a_{e,k} \to |h_e|$, and $\underline{z}_{e,k} \to x_e + h_e$. Strict domain and recovered-positivity margins make every denominator positive for all sufficiently large k. For each such k, equation (29) gives $\Gamma_{U,k} \ge G(y,T)$, while continuity of the finite sum gives

$$\lim_{k\to\infty} \Gamma_{U,k} = G(y,T) = \sum_{e\in T} \frac{\mu_e h_e^2}{x_e^2(x_e+h_e)}.$$

No nesting or monotonicity of the upper endpoints is required. If $G(y,T) < \varepsilon$, the final comparison passes for every sufficiently large k. This proves proposition 4.1.

For comparison, the legacy sign-oblivious predicate replaces the signed lower bound on $x + h$ by $x - |h|$. If $\rho = \max_e |h_e|/x_e \ge 1$, its recovered-positivity gate rejects and we set $\Gamma_{\text{abs}} = +\infty$; no finite limiting comparison is claimed. If $\rho < 1$, its bound converges to Γ_{abs} in equation (15). On an edge with $h < 0$ the legacy and exact denominators coincide. For $h > 0$, their ratio is $(x+h)/(x-h)$. Edges with $h = 0$ contribute zero to both sums, so the termwise inequality holds trivially there. Hence every nonzero term ratio lies in $[1, (1+\rho)/(1-\rho)]$. Summing nonnegative terms proves equation (16). This is an ablation result for the retired absolute predicate, not a limitation of the production signed predicate.

Binary64 realization. The implementation additionally computes

$$\widehat{z} = \mathrm{fl}(\mathrm{fl}(x(y)) + \mathrm{fl}(h_T(y))) \tag{36}$$

for downstream use. Its positivity and residual are measured, but its rounded entries are not substituted into the exact-real theorem. The certificate binds $Z(P,y,T)$, not a claim that $B\widehat{z} = \beta$ bitwise.

The same distinction is essential in the MMFL interpretation. Theorem 2.1 applies only when the probabilities in its Horvitz–Thompson denominators equal the actual categorical inclusion probabilities. A sampler that row-normalizes $\widehat{z}$, overwrites a final CDF endpoint, or otherwise changes the realized inclusion law cannot continue to use the unmodified entries of $\widehat{z}$ as exact denominators without introducing bias. A rigorous downstream interface must therefore either sample from the implicit object using a certified exact-real comparison procedure, or define and use one finite realized probability vector consistently in both the sampler and estimator while separately certifying or reporting its marginal and objective perturbations. The measured residual of $\widehat{z}$ alone does not upgrade it to the exact-real object authorized by the endpoint theorem. Accordingly, a conditional-MSE value evaluated in binary64 at $\widehat{z}$ is a diagnostic for that materialized probability vector; it is not an interval consequence of the certificate for Z. The experiments keep strict coverage, materialized residuals, and materialized conditional MSE as separate fields.

D Linear verifier and lower-bound details

For each client and model group, preprocessing stores a stable permutation, segment boundaries, and the balanced pairwise-addition schedule. The tree plan stores level order, active parents, child signs, compact group labels, and one pairwise schedule per level. With a fixed topology these arrays occupy $O(m+n)$ words.

Each grouped-reduction plan uses an active-frontier DAG. Within one plan, a nonempty group of degree d creates exactly $d-1$ addition nodes: leaves that have no partner are carried only as frontier references, and a completed group leaves the frontier. Thus one plan with g nonempty groups over m inputs creates exactly

$$\sum_{j=1}^{g}(d_j-1)=m-g \tag{37}$$

stored additions and at most $2m-g$ workspace values including the inputs. This is a per-plan count, not a count summed over every verifier plan. It is independent of $d_{\max}$, including one giant group together with $\Theta(m)$ singleton groups. Applying the same argument separately to the incidence plans and to each disjoint tree-level grouping gives total topology storage $O(m+n)$ and total tree-plan storage $O(n)$.

One recovery-only replay performs a constant number of edgewise interval operations, two grouped incidence reductions, a tree pass in which every vertex and tree edge is visited a constant number of times, and one gap reduction over tree edges. Hence the arithmetic work and temporary state are $O(m+n)$. A connected graph has $m \geq n-1$, so both are $O(m)$.

For a restricted probe observation, consider zero-error verifiers whose accepting authority explicitly establishes every dual inequality $c_e - b_e^\top y > 0$ by reading fresh, explicit cost arrays. No trusted digest, proof oracle, or certified dynamic-update oracle is part of this input contract. Fix the legal one-word cost alphabet $\mathbb{W}$ and the represented input class. Relative to an accepting trace and its candidate y, call edge e *slack-sensitive* if there exists $\widetilde{c}_e \in \mathbb{W}$ such that replacing only c_e by $\widetilde{c}_e$ leaves a legal represented input and

$$\widetilde{c}_e - b_e^\top y \leq 0. \tag{38}$$

This condition is explicit: it is not automatic under a bounded binary64 range, a nonnegative-cost restriction, or an overflowed potential difference.

Now take an accepting trace of a zero-error verifier in this restricted class and fix any random coins. If the trace omits a slack-sensitive word c_{e_0}, perform the legal replacement from equation (38). Every probed word, branch, and proposed auxiliary certificate remains unchanged, so the verifier still accepts although the claimed strict slack is nonpositive, contradicting soundness. Thus every accepting trace probes every slack-sensitive cost coordinate. Under the additional premise that all m edges are slack-sensitive, this gives $\Omega(m)$ word probes. This observation is not a lower bound for general certificate systems: it says nothing about an always-reject algorithm, authenticated preprocessing, a proof oracle, or an input class in which some edge can never cross the slack boundary. It also does not apply to a sound primal certificate that establishes optimality without individually proving these dual slacks. Accordingly this is a lower bound for fresh-array, edgewise dual-slack verification, not for every possible endpoint verifier.

E Exact-real local recurrences and executable scope

All pseudoinverse inequalities below are restricted to $\mathbf{1}^\perp$. The case $\lambda = 0$ is stationary because positive definiteness on $\mathbf{1}^\perp$ implies $g = 0$; all contraction ratios below assume $\lambda > 0$. For $v \in \mathbf{1}^\perp$ and balanced r, define

$$\|v\|_H := \sqrt{v^\top H v}, \qquad \|r\|_{H^\dagger} := \sqrt{r^\top H^\dagger r}. \tag{39}$$

At the current point let

$$s = -H^\dagger g, \qquad \lambda = \|s\|_H, \qquad \sigma = \min_e \mu_e^{1/4} q_e^{1/4}, \qquad \eta = \frac{\sqrt{2}\lambda}{\sigma}. \tag{40}$$

For any direction d whose segment remains in the dual domain, set $y_+ = y + d$ and define $q_+, g_+, H_+, \sigma_+, \lambda_+$ at y_+ by the same formulas, with

$$\lambda_+ := \sqrt{g_+^\top H_+^\dagger g_+}, \qquad \eta_+(d) := \frac{\sqrt{2}\lambda_+}{\sigma_+}. \tag{41}$$

The following lemma isolates the one estimate used by all three local tiers. It applies to an arbitrary exact-real direction, rather than only to a direction produced by a particular linear solver.

Lemma E.1 (Unified inexact-direction bound) *Let $d \in \mathbf{1}^\perp$ and define*

$$r(d) := g + Hd, \qquad \theta(d) := \frac{\|r(d)\|_{H^\dagger}}{\lambda}, \qquad \kappa(d) := \frac{\|d\|_H}{\lambda}, \qquad \rho(d) := \max_e \frac{|b_e^\top d|}{q_e}. \tag{42}$$

If $\rho(d) < 1$, every point $y + td$, $t \in [0,1]$, is in the dual domain and

$$\begin{aligned} &\eta_+(d) \le \eta A(\rho(d)) \left[\theta(d) + \left((1-\rho(d))^{-3/2} - 1\right)\kappa(d)\right], \\ &A(\rho) := (1-\rho)^{-1/4}(1+\rho)^{3/4}. \end{aligned} \tag{43}$$

Moreover, $\rho(d) \le \kappa(d)\eta$.

Proof For edge incidence b_e and $H = BWB^\top$, Thomson's principle gives

$$b_e^\top H^\dagger b_e \le W_e^{-1}, \tag{44}$$

because sending one unit only on edge e is a feasible comparison flow for the demand b_e. Energy Cauchy–Schwarz and $W_e = \sqrt{\mu_e}/(2q_e^{3/2})$ therefore imply

$$\frac{|b_e^\top d|}{q_e} \le \frac{\|d\|_H}{q_e\sqrt{W_e}} = \frac{\sqrt{2}\,\|d\|_H}{\mu_e^{1/4} q_e^{1/4}} \le \kappa(d)\eta. \tag{45}$$

This proves the final assertion. If $\rho := \rho(d) < 1$, then $q_e(y+td) = q_e - tb_e^\top d \ge (1-\rho)q_e > 0$ for $0 \le t \le 1$, so the entire segment is in the domain. With $H_t := H(y+td)$, the corresponding edge-weight ratios give

$$(1+\rho)^{-3/2} H \preceq H_t \preceq (1-\rho)^{-3/2} H. \tag{46}$$

Since $g(y+d) = g + \int_0^1 H_t d\, dt$, adding and subtracting Hd yields

$$g_+ = r(d) + \int_0^1 (H_t - H) d\, dt. \tag{47}$$

On $\mathbf{1}^\perp$, equation (46) bounds the H-scaled operator norm of $H_t - H$ by $(1-\rho)^{-3/2} - 1$. Hence

$$\|g_+\|_{H^\dagger} \le \lambda \left[\theta(d) + \left((1-\rho)^{-3/2} - 1\right)\kappa(d)\right]. \tag{48}$$

At the endpoint, $H_+ \succeq (1+\rho)^{-3/2} H$ and $\sigma_+ \ge (1-\rho)^{1/4}\sigma$. Thus $\|g_+\|_{H_+^\dagger} \le (1+\rho)^{3/4}\|g_+\|_{H^\dagger}$; substituting this and equation (48) into $\eta_+(d) = \sqrt{2}\,\|g_+\|_{H_+^\dagger}/\sigma_+$ proves equation (43).

For the exact Newton direction $d = s$, one has $\theta(s) = 0$, $\kappa(s) = 1$, and

$$\rho_s := \rho(s) \le \eta. \tag{49}$$

Consequently, when $\eta \le 0.1$, the entire segment is in the domain and lemma E.1 gives

$$\eta_+(s) \le \eta(1-\rho_s)^{-1/4}(1+\rho_s)^{3/4}\left[(1-\rho_s)^{-3/2} - 1\right]. \tag{50}$$

Direct differentiation shows that the scalar factor multiplying η, divided by ρ_s, is increasing on $[0, 0.1]$ and is below 1.889 at the endpoint. Together with $\rho_s \le \eta$, this proves $\eta_+(s) \le 2\eta^2$.

Continue to assume $\eta \le 0.1$. For an inexact direction $\widetilde{s} = s+e \in \mathbf{1}^\perp$ with Newton residual $r_N = H\widetilde{s} + g$, one has $r_N = He$ and therefore $\|e\|_H = \|r_N\|_{H^\dagger}$. Writing $\theta := \|r_N\|_{H^\dagger}/\lambda$, the triangle inequality gives $\kappa(\widetilde{s}) \le 1+\theta$ and

$$\rho_{\widetilde{s}} := \rho(\widetilde{s}) \le (1+\theta)\eta. \tag{51}$$

Under the exact mathematical forcing condition

$$\|r_N\|_{H^\dagger} \le \frac{\eta\lambda}{4}, \tag{52}$$

one has $\theta \le \eta/4$ and $\rho_{\widetilde{s}} \le (1+\eta/4)\eta \le 0.1025$. The entire segment is therefore in the domain, and lemma E.1 gives

$$\begin{aligned} \eta_+(\widetilde{s}) \le \eta(1-\rho_{\widetilde{s}})^{-1/4}(1+\rho_{\widetilde{s}})^{3/4} \\ \times \left[\frac{\eta}{4} + \left((1-\rho_{\widetilde{s}})^{-3/2} - 1\right)(1+\eta/4)\right] < 3\eta^2. \end{aligned} \tag{53}$$

The ratio of the displayed upper bound to η^2 is increasing for $0 < \eta \le 0.1$ after substituting $\rho_{\widetilde{s}} = (1+\eta/4)\eta$; its endpoint is below 2.272.

There is also a weaker exact-real contraction tier. If $\theta \le 0.4$, then equation (51) gives $\rho_{\widetilde{s}} \le (1+\theta)\eta \le 0.14$, so the entire segment remains in the domain. The unified bound specializes to

$$\eta_+(\widetilde{s}) \le \eta A(\rho_{\widetilde{s}}) \left[\theta + \left((1-\rho_{\widetilde{s}})^{-3/2} - 1\right)(1+\theta)\right]. \tag{54}$$

Every displayed factor is nondecreasing on the stated rectangle. At $\rho = 7/50$,

$$A(7/50)^4 = \frac{57^3}{50^2\, 43} < \left(\frac{573}{500}\right)^4.$$

Moreover,

$$\left(\frac{50}{43}\right)^3 < \left(\frac{3511}{2800}\right)^2,$$

and hence

$$\frac{2}{5} + \left((43/50)^{-3/2} - 1\right)\frac{7}{5} = \frac{7}{5}\left(\frac{50}{43}\right)^{3/2} - 1 < \frac{1511}{2000}.$$

Therefore

$$A(7/50)\left[\frac{2}{5} + \left((43/50)^{-3/2} - 1\right)\frac{7}{5}\right] < \frac{573}{500}\frac{1511}{2000} = \frac{865803}{10^6} < \frac{433}{500}.$$

Hence $\eta_+(\widetilde{s}) < 0.866\eta \le 0.0866$ and the exact-real basin is invariant. A sequence using this tier has linear contraction; only the quadratic tier in equation (53) supports the corresponding doubly logarithmic local iteration count.

The production screens obtain their energy bounds without solving another Laplacian system. For any balanced demand d, Thomson's principle gives

$$\|d\|_{H^\dagger}^2 = \min_{Bf=d} \sum_e \frac{f_e^2}{W_e} \le \sum_{e\in T} \frac{f_{T,e}(d)^2}{W_e}. \tag{55}$$

Signed interval tree routing encloses $f_T(d)$, and outward lower bounds on W_e therefore give an executable upper bound on the right-hand side. Applied to $d = g$, this gives the basin upper bound on η; applied to $d = r_N$, its square root is the forcing quantity $R_U \ge \|r_N\|_{H^\dagger}$.

For the lower comparison, define the exact quantity

$$\lambda_L := \begin{cases} |g^\top \widetilde{s}| / \|\widetilde{s}\|_H, & \|\widetilde{s}\|_H > 0, \\ 0, & \|\widetilde{s}\|_H = 0. \end{cases}$$

Cauchy–Schwarz gives $\lambda_L \le \lambda$. The executable replay evaluates the reduced-gauge pairing with a directed absolute lower endpoint and evaluates $\|\widetilde{s}\|_H$ from an outward upper energy endpoint. Their quotient is a lower bound $\underline{\lambda}_L \le \lambda_L$. Thus $R_U \le 0.4\underline{\lambda}_L$ implies the exact-real contraction hypothesis. However, the implementation evaluates this screen for the proposed binary64 direction and then stores a separately rounded update, including a rounded gauge normalization. The screen therefore does not certify that realized stored transition. CertQuota-S is described as interval-screened and terminal-endpoint-certified, not as an executable path certificate. Endpoint soundness in theorem 4.1 assumes none of these local conditions.

F Scaled standard-SOC candidate and dual recovery

The conic reference uses $t_e \ge 1/x_e$. Choose a positive flow scale s_x and write

$$x = s_x z, \qquad t = s_x^{-1} w. \tag{56}$$

Then $x_e t_e \ge 1$ is exactly $z_e w_e \ge 1$, which is equivalent to the standard second-order cone

$$\|(2, z_e - w_e)\|_2 \le z_e + w_e. \tag{57}$$

The scaled objective is

$$\frac{c^\top (s_x z) + \mu^\top (s_x^{-1} w)}{s_f}, \tag{58}$$

where $s_f > 0$ is an objective scale. Let $R = \operatorname{diag}(r_1, \dots, r_{n-1})$ be the row scaling and impose

$$R\widetilde{B}(s_x z) = R\widetilde{\beta}. \tag{59}$$

If CVXPY/Clarabel returns equality multiplier ξ for left-minus-right zero, stationarity with respect to z gives

$$\frac{1}{s_f}\left(c - \frac{\mu}{x^2}\right) + \widetilde{B}^\top R\xi = 0. \tag{60}$$

Since the reciprocal KKT convention is $c - \mu/x^2 = B^\top y$, the canonical reduced potential is

$$y_{1:n-1} = -s_f R\xi, \qquad y_n = 0, \tag{61}$$

followed by an optional mean-centering gauge shift. Both the objective scaling and the minus sign in equation (61) are essential. The conic solver's own status is not trusted: the recovered y is passed to IEEE-V.

G Finite-precision and caching contract

The IEEE implementation relies on round-to-nearest binary64 operations plus directed predecessor/successor adjustments to enclose exact dyadic expressions. Its fail-closed platform gate executes six classes of spot probes: binary64 layout, boundary behavior of `nextafter`, ties-to-even, gradual underflow versus FTZ/DAZ semantics, selected $+, -, \times, /$ identities, and selected square roots. Successful static probes are cached, but lightweight current-thread sentinels are repeated at every certificate entry and when a prepared state is reused. Thus a detected rounding-mode or subnormal-control change causes rejection. These finite probes are not a formal proof of correct rounding over all operands, all threads and future states, every SIMD or compiler path, or the entire square-root domain; those remain the explicit hypotheses of theorem C.1 and lemma C.1. Probe success therefore does not discharge the arithmetic assumption; it only permits the fail-closed implementation to proceed under that assumption. High-precision Arb evaluation is called only by the audit backend, never by the timed IEEE path.

The cache has two ownership levels. A problem topology owns client/model grouped-reduction plans, and the fixed tree owns its level-reduction plans. A dynamic problem may reuse a topology owner only after exact endpoint-array and dimension checks. The cache contains no q, x, g, tree-flow, positivity, or gap endpoint. In the strict frontier, common topology and tree plans are prewarmed before the method-order rotation; the current-identity plan needed by legacy stepwise code is likewise prewarmed before each round. These times are reported separately and are not charged to whichever method happens to run first.

The strict certificate object stores the backend, requested-tolerance binding, an outward upper bound on the pre-correction $\|g(y)\|_1$, a lower bound on $\min_e x_e(y)$ before correction, the gap upper bound, acceptance flags, and content fingerprints for its replay inputs. It intentionally does not retain the full edgewise $\{\ell_e, a_e\}$ arrays. Authority is process-local: an unsealed fabricated summary cannot authorize a return, and a serialized summary must be replayed from its bound inputs rather than treated as a self-contained proof. The complete in-process representation is the immutable replay tuple

$$(P, y, T, \text{certificate}), \tag{62}$$

which identifies $Z(P, y, T)$. The separately emitted binary64 vector and measured residual are convenience outputs, not replacements for this representation.

H A42 confirmatory and audit protocol

Release boundary and evidence graph. The full-paper release identifier is `certquota-fullpaper-v1-a42`. A42 is a clean computation release, not an incremental update to A41: no A41 raw row, timing, aggregate, bootstrap draw, or figure may enter an A42 result. A41 remains immutable historical evidence. After code, tests, the frozen protocol, and fixed configuration files are closed, one execution-source manifest is created exactly once; every formal process rehashes that closure and binds the same manifest digest before doing work.

The evidence graph contains 13 logical producers. Nine rerun inherited frozen designs: certificate behavior, dynamic sequences, the named-method strict frontier, internal standard-SOC reference, external standard-SOC stress, verifier scaling, conditional-MSE allocation, official MovieLens topology, and the independent Arb audit. The last producer has a 256-bit primary pass and a 384-bit rejection-safety pass but remains one logical panel. Four new producers are S1 (skewed reduction), S2 (certificate sharpness), A1 (MovieLens user–genre sampling), and P1 (full solve–verify–materialize scaling). Twelve analysis products are then created: strict-frontier and internal-SOC records share one analysis, while the two Arb passes share another; all other panels have one analysis each. This declared count prevents an unregistered result or a selectively omitted panel from entering the manuscript.

Clean-execution contract. Every target and its exclusive `.summary.json` sidecar must be absent at launch. Producers run sequentially with the five OpenMP/BLAS thread variables fixed to one and with the source-manifest path and digest in the environment. Resume, sharding, hand-selected cell indices, resource-dependent substitution, and pooling across releases are forbidden. A sidecar binds the raw file's repository-relative path, SHA-256, exact nonblank row count, execution-source digest, and runtime contract; failures and non-success statuses remain in the denominator. Analyses read only these A42-bound files. Numerical statements in the paper are generated from the 12 analyses through one macro file, rather than transcribed manually.

Inherited dynamic grid. For each of three graph families and seeds 3701–3705, the design uses two 20-round smooth sequences at perturbation scales 10^{-3} and 10^{-2} and one 30-round shock sequence. The resulting 45 sequences contain 1,050 rounds. The order of CertQuota-V, CertQuota-S, and damped Newton is offset by $(\text{cell} + \text{round} - 1) \bmod 3$, giving exactly 350 appearances in each method position; ratios are paired within round. The bootstrap resamples the 15 independent (graph family, sequence-seed) clusters, with scales and smooth/shock trajectories treated as subsequences within cluster. Basin-screen membership and terminal endpoint acceptance are reported separately. This is a smooth/shock warm-start experiment, not an arbitrary-initialization global solver comparison.

Inherited named-method strict frontier. The frontier crosses the client-regular, community, and bottleneck families; seeds 3711–3715; 100,000 edges; five rounds; and perturbation scale 10^{-3}. The four order-rotated methods are CertQuota-V, the frozen interval-screened IEEE candidate pipeline followed by terminal IEEE replay (CertQuota-S), the same candidate cascade followed by 70-digit Arb replay, and alternate scaling followed by the same IEEE replay. Total method time begins with candidate generation and includes strict-state preparation, authoritative verification, and materialization; common router and topology preparation are reported separately. Candidate status and long-tail timing quantiles are retained. The panel compares named variants under one representation, process, software, and hardware contract. Its ranking field is fixed to `claim_eligible=false`; it cannot authorize a fastest-solver claim.

Inherited standard-SOC reference and stress panels. The internal correctness grid crosses three families, five seeds, and edge targets 300, 1,000, and 3,000, giving 45 instances and 90 method records. Both the CertQuota-V and CVXPY/Clarabel standard-SOC candidates are submitted to the same terminal IEEE endpoint verifier. This panel can establish small-instance agreement only; it is not an external timing ranking.

The external stress grid crosses 10,000 and 30,000 edges, the same three families, and seeds 3811–3813, giving 18 instances and 36 method records. All candidate failures and verifier rejections are retained. Reference and performance gates require complete endpoint-authorized matching; if either gate fails, no external speedup is estimated. The optional 100,000-edge case is not part of the formal grid. Environment probes record unavailable packages such as VSDP or BlockIP as unavailable, never as synthetic timeouts.

Inherited independent Arb audit. The 256-bit implementation imports none of the production solver, certificate, interval, tree, or verifier modules. From frozen public arrays it independently reconstructs the standard-SOC candidate convention, equality-dual sign and scaling, stable-Kruskal tree, exact-dyadic tree routing, and stable recovery-gap sum. Deterministic replay is bound by graph and primal/dual array hashes, iteration counts, and frozen candidate metrics. A second 384-bit pass revisits every primary-pass rejection named by the registered design. The checker may diagnose agreement or bound inflation for the same predicate; it is neither an external baseline nor a general verified conic solver.

Inherited verifier scaling. Edge targets 30,000, 100,000, 300,000, one million, three million, and ten million are crossed with seeds 3701–3710. One warm-up is excluded and three timed endpoint replays are retained per instance. A seed-cluster bootstrap fits log time on log edge count. Recursive storage accounting follows unique NumPy base chains and memoryview owners, counts shared owners once across problem and tree plans, and excludes the weak-reference cache, transient compilation storage, and allocator overhead. Complete grid coverage and the structural storage gate are primary; fitted runtime exponents are descriptive, not proofs of complexity.

Inherited conditional-MSE allocation grid. The 32 graph cells cross client counts 100 and 1,000; model counts 3 and 10; eligibility degrees 2 and 5 capped by model count; Gamma-shape heterogeneity 0.1 and 1; and quota ratios 1 and 10. Each cell has five independent graph/update replicates, each with 20 unreported burn-in fields followed by 20 evaluation fields and 11 policies: one analytic uniform policy and ten nonuniform policies. The design therefore specifies 35,200 records, including 32,000 strict solves. Stale ages 1, 5, and 20 must use the declared past norm field without fallback. MSE ratios are paired within evaluation field, inference clusters by the 160 graph replicates, and Holm's procedure adjusts the two primary comparisons; effects are also stratified by quota ratio. The endpoint is the analytic conditional-MSE formula evaluated at the materialized binary64 probability vector for the stated independent categorical law, not an interval certificate for a pseudorandom sampler implementation or a neural-training result. The registered row targets satisfy $\max_i r_i \leq 0.2$; the materialized residual gate therefore keeps every row mass below one, and the same positive finite vector is used as both the categorical law and the Horvitz–Thompson denominator.

Inherited official-topology stress. The A39 design parses the official MovieLens-1M archive identified by MD5 `c4d9eecfca2ab87c1945afe126590906`; the archive is neither downloaded by the run nor redistributed. User–movie eligibility is observational, while weights, costs, marginals, optimum, and warm starts are synthetic. Five frozen planted-weight cases are submitted to strict endpoint verification. This topology stress can support scale and provenance claims, but neither recommender quality nor a solver ranking [24].

Inherited untimed certificate behavior. The frozen design crosses three graph families; reciprocal-weight spans 0, 4, and 8; quota ratios 1 and 10; and valid, cold, and deliberately domain-invalid potentials, giving 54 planned cells. Direct verification and terminal control flow are both recorded, including repair and explicit failure. Every planned cell remains in the artifact. This tests fail-closed rejection and control-flow coverage, not runtime superiority.

S1: adversarial segmented-reduction topology. For each

$$m \in \{30{,}000, 100{,}000, 300{,}000, 10^6, 3 \cdot 10^6, 10^7\},$$

one group contains half the inputs and every remaining input is a singleton. Ten fixed seeds 42001–42010 generate finite positive binary64 values. The plan is compiled once; one execution is excluded as warm-up and three are timed. The primary gate is structural: with g nonempty groups, the stored addition-node count must equal $m - g$, workspace length must equal $m + (m - g)$, plan bytes must obey the declared linear dtype bound, and no allocation or arithmetic substitution is allowed. Outputs are bitwise compared with an independently executed stable levelwise reference through 300,000 inputs and checked against an analytic/high-precision enclosure above that size. Timing slopes are descriptive; the exact node count, not a regression, validates the storage/work construction.

S2: certificate sharpness and acceptance envelope. Thirty public-array base instances cross the three graph families with scales $(64, 8, 4)$, $(256, 32, 4)$, and $(1024, 128, 4)$ for (clients, models, degree), using respectively four, four, and two replicates per family. A damped-Newton solve that cannot access planted optima supplies each reference dual point. Seven seeded relative-slack perturbation radii $\{0, 10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}, 5 \cdot 10^{-2}\}$ produce 210 candidates; decisions are evaluated at five frozen relative tolerances from 10^{-14} through 10^{-6}.

For every candidate on the same fixed tree, a 100-decimal independent replay computes the exact recovery identity gap G, the limiting absolute-correction bound $\mathit{\Gamma}_{\text{abs}}$, the production signed outward bound $\mathit{\Gamma}_U$, the signed edgewise decision, and the legacy global-ℓ_1 decision. For every finite positive case the primary gate requires

$$G \leq \mathit{\Gamma}_{\text{abs}} \leq \frac{1+\rho}{1-\rho} G, \qquad \rho = \max_e |h_e|/x_e < 1,$$

requires Γ_U to contain G under the declared binary64 contract, and requires the signed upper bound not to exceed the absolute-predicate upper bound when both replay identical enclosures. Domain and positivity rejections remain in the denominator. Inflation and acceptance differences are descriptive measures of sharpness, not new soundness assumptions.

A1: MovieLens user–genre sampling case study. This panel reuses the same checksum-verified local MovieLens-1M archive and reads only ratings and movie genres, not demographic fields. Genres are model tasks and users are clients. The common support is selected retrospectively by requiring positive observations on both sides of each cutoff. Cutoff quantiles $\{0.40, 0.45, 0.50, 0.55, 0.60\}$ create historical/later splits on common user–genre support. Rating-derived two-coordinate update proxies have a strictly positive second coordinate on every retained edge. Each cutoff is crossed with equal and history-popularity model quotas and with three quota-matched policies: later-update norm (a diagnostic oracle), historical update norm (a past-only weight proxy evaluated on that retrospective support), and equal edge weight. Every optimized policy must pass the same endpoint verifier. The endpoint is analytic conditional Horvitz–Thompson MSE on later updates. The ten paired cells are a descriptive allocation case study; they authorize no neural training, recommendation, demographic-fairness, or held-out-ranking claim.

P1: full solve–verify–materialize scaling. The three graph families are crossed with edge targets 30,000, 100,000, 300,000, one million, and three million and seeds 42501–42503, yielding 45 setup cells. All use degree 8, reciprocal-weight log span 4, quota ratio 1, zero linear cost, and maximum client budget 0.2. A deterministic seeded perturbation of the planted dual optimum has maximum relative slack change exactly 0.05. Graph generation, router construction, and eager plan compilation are measured as cold setup but excluded from the timed pipeline. After one excluded warm-up, three repetitions per cell time the public `solve_verified` call through returned binary64 materialization, giving 135 planned timing records. Candidate, prepare, verify, fallback, materialization, and unattributed phase times and every attempted-candidate status remain visible.

The primary P1 gate requires exact 45-cell and 135-record coverage, strict endpoint acceptance for every warm-up and timed repetition, finite numeric fields, normalized materialized marginal residual at most 10^{-9}, and exact raw/sidecar/source binding. Seed-median timings, phase shares, and log–log slopes are descriptive. The panel closes the verifier-only scope gap, but its regression does not prove a globally near-linear candidate solver.

A43 reviewer-obstacle amendment. A43 is frozen separately under `certquota-reviewer-obstacles-a43`; its execution-source SHA-256 begins `e78292a27473` and binds 139 files. It neither changes nor pools any A42 population. T1 crosses three graph families, two scales, three seeds, two registered candidate perturbations, four tree rules, and four objective-scaled tolerances. The same public-array damped-Newton candidate is replayed on every tree. All decisions, finite bounds, correction magnitudes, setup times, and verification times remain in the 576-row denominator. Tree insensitivity is authorized only if all 144 cells agree and the registered 95th percentile finite-gap ratio is at most 1.10; the observed panel fails both conditions and is reported as a negative result.

R1 uses cutoff quantiles 0.40, 0.50, and 0.60 of the checksum-verified MovieLens timestamps. Support, pair counts, rating coordinates, reciprocal weights, induced quotas, public start, and maximum-conductance tree use ratings at or before the cutoff only. Pairs need at least two historical ratings and the largest connected historical component is retained. The resulting quotas are induced by a strictly positive within-user historical-count flow, so no planted optimum is consulted. The endpoint is optimization acceptance and a materialized residual; recommendation accuracy is ineligible.

One accepted R1 result is serialized as a canonical `certquota-endpoint-replay-v1` bundle. It stores exact binary64 arrays, candidate, tree, tolerance, and a SHA-256 payload binding, but no serializable authority. A fresh object reconstruction reruns the strict backend and must both accept and match the claimed fingerprints. Tampering, malformed arrays, and unaccepted exports fail closed in tests. The first R1 CLI attempt used a relative output argument and failed while writing its summary after all solves; its unchanged raw/replay files are retained under a named quarantine and are ineligible. The panel was rerun from zero under the same frozen source hash with an absolute output argument; only the complete exclusive-create rerun is analyzed.

Release authorization and claim boundary. Eight PDF/PNG figure pairs are regenerated from the 12 A42 analyses. Only after the manuscript and all derivatives are final is a submission manifest created. The read-only release audit recomputes every analysis, reparses both MovieLens panels from the local official archive, byte-compares all eight figure pairs, builds one wheel and one source distribution in an isolated copy, runs package metadata checks, requires the full test suite to pass with zero skips, and performs an isolated recorder-checked LaTeX build. A missing or skipped component produces a partial, non-authorizing report.

The A43 analyzer separately rehashes its execution closure, raw files, sidecars, and replay; recomputes all tree disagreement and gap-ratio fields; checks every historical-input flag and official archive checksum; and reruns the portable endpoint checker. Its submission manifest then binds the A42 historical closure, A43 evidence, revised manuscript, and compiled PDF. The A43 evidence audit deeply compares the recomputation and requires a full zero-skip test run. Author identities, declarations, public repository, hosted CI logs, and archival identifiers are distinct from that historical numerical audit. The later author-identified preprint and its public CI checks are described in the software-availability section; no DOI is claimed.

Passing this graph can authorize the represented problem's linear-work endpoint verifier, the signed-certificate soundness and sharpness evidence, and measured throughput for explicitly named implementations under the recorded contract. It cannot authorize novelty of the reciprocal objective, a globally near-linear candidate solver, a universal fastest verified optimizer, MMFL training SOTA, or venue acceptance. Internal SOC agreement, external completeness, independent-predicate replay, and application evidence are separate gates and are not substituted for one another.

I A45 independent witnesses and exact output contract

The A45 checker is a separate Python-standard-library implementation. It uses intervals $[a, b]2^{-160}$ with integer endpoints. Addition and negation are exact; products and quotients are rounded using integer floor/ceiling division; square-root endpoints use integer square root and an upward increment unless the upper radicand is a perfect square. A zero-containing divisor, nonpositive certified flow, malformed graph, changed payload, or unsupported dimension rejects. The 200,000-edge input limit is an audit-tool resource bound, not a claim about the production verifier. Python big-integer correctness remains part of its trusted computing base; this is not a formal proof assistant.

For endpoint input, the checker reconstructs every binary64 number as its exact rational value, derives the distinguished marginal by exact balance, validates the supplied tree by its own adjacency traversal, and encloses the reciprocal KKT flow and leaf-elimination correction. It directly encloses $P(Z)$ and $D(y)$ and compares $P(Z)^+ - D(y)^-$ with the exact represented tolerance. It does not trust a serialized production gap, solver status, production capability token, or production interval operation. The audit may be more conservative than a symbolic cancellation bound and is not part of the binary64 linear-work timing theorem.

For conic-port witnesses, let s_x, s_o and R be the positive dyadic primal, objective and row scales. With lifted variables (z, w, s, u, v), the equations are $RB(s_x z) = R\beta$, $u = 2$, $s = z + w$, $v = z - w$ and the objective is $(s_x c)^\top z/s_o + (\mu/s_x)^\top w/s_o$. The checker reconstructs the dual slack from these equations, verifies each nonnegative-orthant entry and $d_s \geq 0$, $d_s^2 \geq d_u^2 + d_v^2$ using exact rational arithmetic, and proves the claimed lower endpoint by weak duality with scale s_o. No exported lifted matrix or cone-residual summary carries authority.

For the upper endpoint it treats the exported primal midpoint merely as a dyadic candidate, repairs its original marginal residual by rational tree elimination, checks all exact equations and positivity, and encloses its reciprocal objective below the claimed upper endpoint. This proves the original-input bracket even though it is not a proof that every compatibility routine implements all INTLAB semantics. Point-matrix, interval-right-hand-side tests with exact Gaussian-elimination corner solutions are an additional compatibility diagnostic, not the bracket's authority.

The decision companion similarly turns a materialized numerical allocation into a separately checked rational allocation $\widehat{Z}$. Its numerator and denominator arrays explicitly identify the feasible vector consumed by the application. This vector need not equal the original implicit Z; positivity, $B\widehat{Z} = \beta$, and its own primal–dual gap are checked anew. For an input cap C, $P(\widehat{Z})^+ \leq C$ proves a feasible policy below cap, $D(y)^- > C$ proves the cap unattainable, and overlapping intervals return undecided. A sampler still needs a correct realization of these rational probabilities; neither these files nor a normalized binary64 approximation constitute a tested categorical sampler.

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Data and software availability. The public companion repository is `https://github.com/long-0228/certquota`, with the paper and release links at `https://long-0228.github.io/certquota/`. The dated preprint release is identified by the tag `preprint-2026-09-06` and its Git commit. The public package contains the core source, installation tests, independent checkers, and a separately hash-manifested archive of 90 synthetic-problem proofs. The 24 MovieLens-derived proof objects checked locally are not redistributed under the dataset's permission requirement. It is a curated publication snapshot, not the complete historical local reviewer bundle. No archival DOI has yet been assigned. The MovieLens-1M source data are available from GroupLens; the checksum-constrained archive is not redistributed by this artifact.

Use of AI-assisted tools. OpenAI Codex was used to assist with software implementation, experiment orchestration, manuscript restructuring, and consistency checking. It is not an author. The human author is responsible for independently checking every mathematical argument, citation, software change, analysis, and statement in the submitted version, and takes full responsibility for that version.